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# The urban tree of life: quantifying relationships between body size and urban tolerance for more than 30,000 plant and animal species

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## eLife Assessment

This study provides an **important** assessment of how body size influences the occurrence of macro-organisms in urban areas across the globe. Size in most plants, but only some animal families, was positively associated with urban tolerance. The data set is impressive, but the evidence for broad-scale conclusions is **incomplete** due to methodological issues that need to be resolved.

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## Abstract

Urbanization is a major global driver of biodiversity change, with species responses to urban settings ranging from avoidance to exploitation. To better understand these responses, we conducted a global analysis of urban tolerance across more than 30,000 animal and plant species, with a focus on body size—an ecological trait fundamental to space use, mobility, metabolism, and environmental sensitivity. Our synthesis showed a consistent pattern across taxa and biogeographic regions: many species are urban avoiders, while few thrive as urban exploiters—a pattern we coin “Species Urbanness Distribution”. We further found that larger body sizes were generally associated with greater urban tolerance in plants compared to animals, though these size-tolerance relationships showed considerable variability among families. Our findings highlight the heterogeneous relationship between body size and urban tolerance across the tree of life, underscoring the importance of tailored strategies to support urban biodiversity. This research advances ecological understanding of urban filtering and provides a framework for guiding biodiversity-sensitive urban planning amid accelerating global urbanization.

## Main

Cities represent novel anthropogenic environments, leading to significant mismatches with the evolutionary history of most organisms<sup>1</sup>. As urban areas are set to expand two- to sixfold over the 21<sup>st</sup> century<sup>2</sup>, urbanization poses a substantial and accelerating threat to global biodiversity<sup>3</sup>. Yet species vary widely in their responses: some are completely extirpated (‘urban avoiders’), others

persist through behavioral or ecological acclimation (‘urban adapters’) and/or rapid adaptive evolution to cities<sup>4,5</sup>, and some even thrive in urban environments<sup>6</sup> (‘urban exploiters’). Quantifying these divergent responses is key for understanding how urbanization will continue to shape global biodiversity and urban ecosystem functioning (e.g., urbanization reshapes a food web<sup>7,8</sup>). However, the young field of urban ecology has typically relied on small-scale studies, and/or has often focused on relatively few species constrained to particular taxonomic groups such as birds<sup>9</sup> or mammals<sup>10</sup>. Thus, whether general patterns of urban preference or avoidance emerge across species within and across regional pools remains untested.

Species likely do not respond randomly to urbanization as traits may modulate species’ responses. Trait-based approaches are powerful for identifying generalizable patterns across diverse taxa, particularly at broad spatial and phylogenetic scales<sup>11</sup>. Although many traits (e.g., diet, reproductive strategy, dispersal ability) may influence species-specific responses to urbanization, body size stands out as both the most widely measured and most taxonomically integrative. It is a key trait of any organism, relating to space use, life-history, and metabolic rate across the tree of life<sup>12,13,14</sup>, making it a plausible direct or indirect predictor of species’ responses to key stressor gradients in urban landscapes, including resource fragmentation, heat stress from urban heat islands (UHI), and anthropogenic disturbance (Fig. 1 [↗](#)).

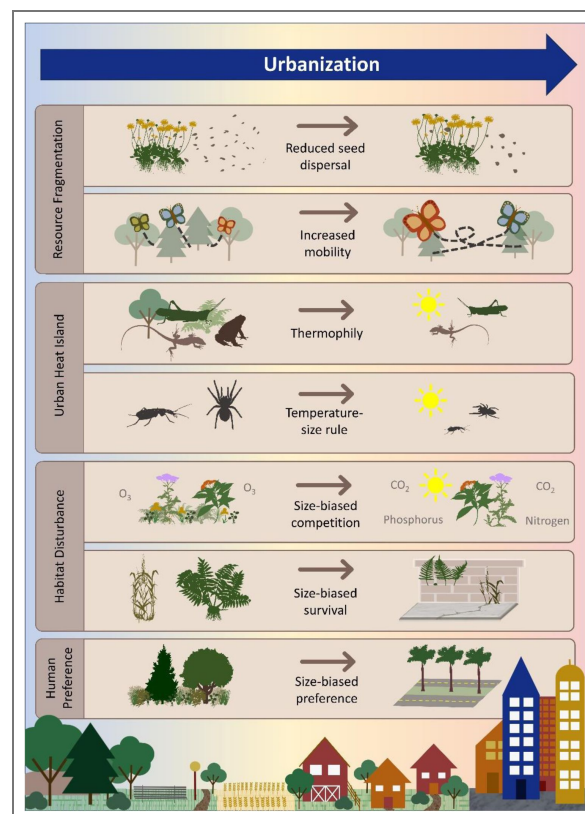
Body size correlates with urban mobility, but in complex ways<sup>15</sup>. For instance, while higher mobility allows some animal species to better cope with the fragmented nature of urban habitat resources, reduced mobility might be advantageous for exploiting localized resources and avoiding risks associated with the urban matrix<sup>16,17</sup>. In plants, body size—often indexed by height—correlates with competitive ability, light acquisition, and reproductive strategy, with larger species generally having greater resource needs but potentially higher resilience to urban stressors<sup>18</sup>. Notably, the UHI may drive shifts to smaller-bodied animal species due to elevated metabolic costs<sup>14,15</sup>, echoing similar patterns observed under global warming<sup>19,20</sup>. However, these shifts toward smaller size can be overruled by requirements for increased urban mobility in taxa where mobility increases with body size<sup>15</sup>. All in all, despite body size’s fundamental role in modulating species responses to their environment, a systematic, cross-taxon assessment of how urbanization filters body size distributions is lacking.

Here, we present a global synthesis of urban tolerance across more than 30,000 animal and plant species, using occurrence records from the Global Biodiversity Information Facility (GBIF) and remotely sensed night-time light intensity as a proxy for urbanization. We define a continuous metric of urban tolerance as a species’ relative affinity to urban areas, with negative values indicating avoidance and positive values indicating preference<sup>21</sup>. To account for regional variation in both species pools and in species’ urban tolerance across their range<sup>22,23</sup>, we calculated urban tolerance within 52 geographic subrealms (Fig. S1 [↗](#)). Our dataset spans 47 taxonomic classes—from flowering plants (8,980 species) to insects (8,674 species) to mammals (648 species)—and represents a uniquely broad assessment across the tree of life. We first introduce the concept of “Species Urbanness Distribution” (SUD) to characterize the composition of a regional community in terms of its species’ urban affinities. We then use Bayesian hierarchical modeling to test whether body size predicts urban tolerance across 17,722 species from 371 families, across multiple subrealms. These analyses allow us to quantify both consistency and heterogeneity of body size effects on urban tolerance across multiple taxonomic levels, and across regions. This synthesis addresses two core questions: (1) What is the shape of species’ urban tolerance distributions within regional communities? (2) Does body size consistently correlate with species’ urban tolerance across taxonomic groups and biogeographic contexts?

## Results and Discussion

### Species Urbanness Distributions (SUDs)

We identified a consistent pattern in the extent to which regional communities—defined as all species within a given biogeographic subrealm—consist of urban avoiders and exploiters. We term this the Species Urbanness Distribution (SUD), which captures the full distribution of urban



**Fig. 1.** We hypothesize a series of mechanisms that could explain interspecific shifts in body size as a function of urban tolerance.

These hypotheses include reduced seed dispersal<sup>59</sup>, increased mobility<sup>42</sup>, thermophily<sup>30</sup>, the temperature-size rule<sup>15</sup>, size-biased competition<sup>79</sup>, size-biased survival<sup>80</sup>, and size-biased human preference<sup>56</sup>.

tolerance values across species in a community. SUDs exhibited a characteristic shape: a pronounced peak at negative values and a long right-skewed tail toward high positive values, indicating that most species are ‘urban avoiders’, while only a few are ‘urban exploiters’. This pattern was consistent across subrealms and several taxonomic levels (Fig. 2). Conceptually, SUDs parallel the well-known Species Abundance Distributions<sup>24</sup> (SADs), which describe the general law of communities being composed of many rare and few common species<sup>24,25</sup>.

Similarly, much like the skewed distributions observed in SADs<sup>24,26</sup>, the skewed shape of SUDs indicates that only a select few species seem to possess traits that enable them to thrive in urban environments. While previous studies have reported a similar pattern at smaller spatial scales<sup>27</sup>, our synthesis demonstrates this applies across the globe.

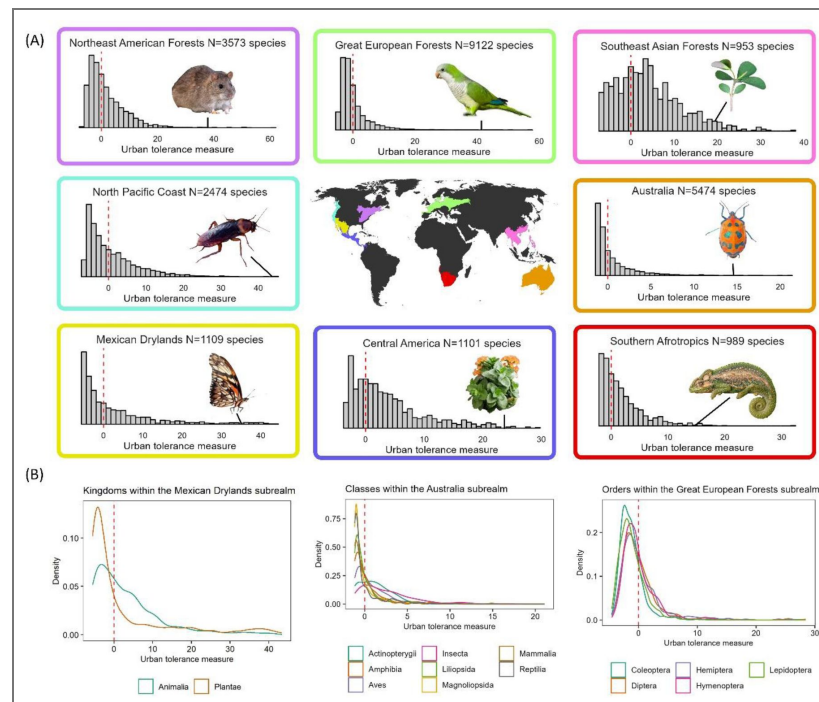
The skewed shape of SUDs suggests that traits enabling species to tolerate urban environments are both rare and specific—an indication that adaptation (*sensu lato*) to urban settings is uncommon, yet occurs geographically widespread across taxa. These traits may be partly found in human-commensal species, which are found globally and often include invasive non-native species<sup>28</sup>. Such traits “pre-adapted” to urban conditions allow for some species to not only persist but thrive in urban environments where most species cannot. Indeed, the median urban tolerance is most often below one, indicating widespread avoidance among species. This is expected, given that urbanization typically involves severe loss and fragmentation of habitat resources for most organisms<sup>1</sup>, which negatively impacts numerous taxa, including birds<sup>29</sup>, beetles<sup>30</sup>, and moths<sup>31</sup>.

Additionally, the UHI—typical of most cities<sup>32</sup>—imposes thermal stress, especially during extreme heat events<sup>33</sup>, which may drive community shifts toward heat-tolerant species, as observed in ants<sup>34</sup>, bees<sup>35</sup>, and plants<sup>36</sup>. Yet, urban environments also frequently contain green infrastructure and remnants of natural habitats that can act as sanctuaries for rare and endangered species<sup>37</sup>, though these spaces are often also dominated by widespread generalist species. The multitude of environmental stressors, combined with competition dynamics between urban-tolerant and intolerant species, likely influence the shape of SUDs (Fig. 2). Future ecological research should aim to disentangle these contrasting drivers across taxa and regions to better understand the ecological filters at play.

## The relationship between body size and urban tolerance

To test whether there is a consistent pattern of body size filtering associated with urban tolerance across the tree of life, we integrated our species-level urban tolerance values with body size measures collated from the literature. We performed this analysis for 17,722 species spanning both plants (i.e., plant height) and animals (e.g., body length, mass, wingspan). We first evaluated the overall relationship between body size and urban tolerance across all taxa, fitting two models (see Methods for details). We found that the effect of body size on urban tolerance is stronger in plants than in animals (0.64 vs. 0.21), although uncertainty was substantial in both groups, with zero-overlapping 95% credible intervals (plants: -0.68 to 1.84; animals: -0.25 to 0.66). Variation in urban tolerance was most pronounced at the family level (sd = 2.43), followed by class and order level (sd = 2.34 and 1.75, respectively). Similarly, the effect of body size varied most among families (sd = 1.24), compared to orders (sd = 0.73) and classes (sd = 0.27), suggesting that body size predicts urban tolerance most consistently within the finer family-level taxonomic scale, rather than at higher scales where other factors may obscure relationships. Based on these patterns, we focused subsequent analyses at the family level, fitting Bayesian models for families with data on at least ten species to independently explore the relationships between urban tolerance and body size.

Across 371 families (93 plant and 278 animal families), we found no universal relationship between body size and urban tolerance; instead, patterns were heterogeneous (Fig. 3). Using 80%, 90%, and 95% credible intervals to infer weak, moderate, and strong effects, respectively, we found evidence of body size filtering in 83 families (29 strong, 19 moderate, and 35 weak). These included 48 animal families (17% of animal families) and 35 plant families (38% of plant families), indicating that for most families, body size does not consistently predict urban tolerance. This suggests that other traits or ecological factors likely underpin the emergence of urban exploiters and the skewed shape of SUDs. Nonetheless, among plant families, the effect of body size on urban



**Fig. 2. Species urbanness distributions (SUDs) exemplified for eight subrealms.**

Plotted are all species per subrealm (A), with the images highlighting an example 'hyper-exploiter' species from each of these subrealms (i.e., with a high urban tolerance score). The x-axis shows the urban tolerance measure whereas the y-axis is the number of species within that bin. There were consistent patterns for kingdoms, classes, and orders (B) as shown by generally similar density histogram shapes for each of these. The vertical dashed line represents where species are neutral towards urban land cover. Photos acquired from iNaturalist CC BY-NC and background was removed by authors: Brown rat (© Ouwesok), Monk parakeet (© Juan Emilio), Cape dwarf chameleon (© Berkeley Lumb), American cockroach (© Len Worthington), Flamingo (© Lyubo Gadzhnev), Junco silverspot (© Rigoberto Ramírez Cortés), Hibiscus harlequin bug (© Sam Fraser-Smith), and Mascarene island leaf-flower (© Douglas Goldman).

tolerance was predominantly ( $N=77$ ; 83%) positive, with 23 families showing strong or moderate effects. In contrast, animal families ( $N=278$ ) exhibited more variable responses: 58% showed positive effects and 42% negative, with only a subset showing strong or moderate effects (positive: 18 families; negative: 6 families). Notable families with the strongest positive associations included Salicaceae (willows) and Columbidae (doves and pigeons), where larger species are more urban-tolerant. In contrast, Dipsadidae (snakes) and Accipitridae (birds of prey) showed the strongest negative associations. Further discussion of notable order-level effects is provided in the Supporting Information. While the overall patterns for family-level effects are visible in Fig. 3 [↗](#), we also developed an interactive online visualization of family-level effect sizes, available [here](#).

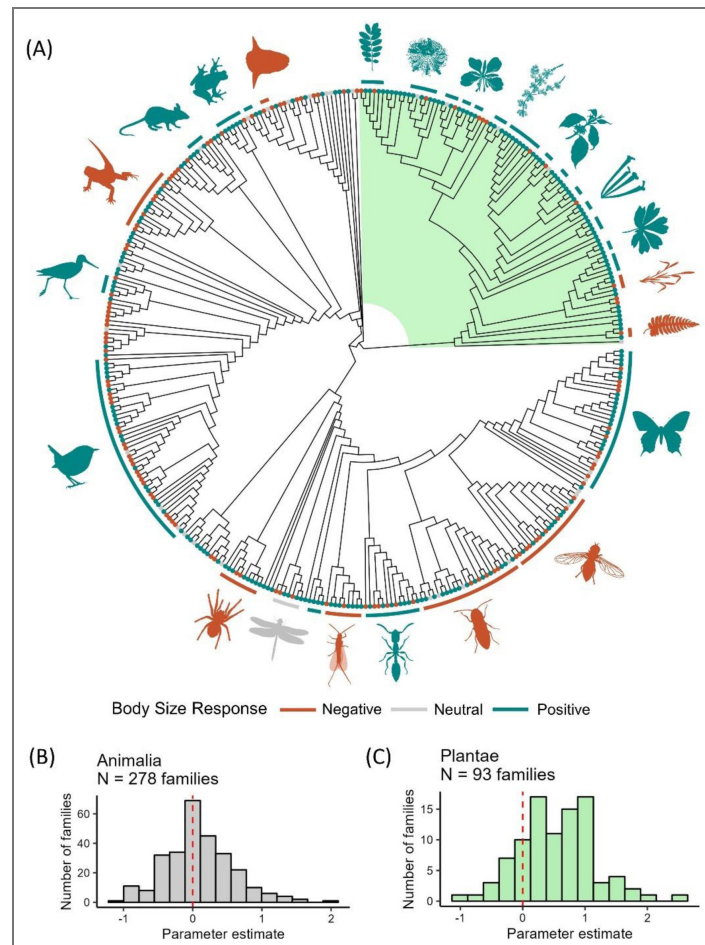
The variability in both the direction and strength of urban body size filtering across taxa likely reflects the diversity of urban stressors and the contrasting taxon-specific pathways through which body size mediates sensitivity to them. In raptors, for example, the negative association between body size and urban tolerance—consistent with previous findings<sup>38,39</sup>—likely stems from large species' need for extensive hunting territories, whose availability and access are constrained by urbanization-driven habitat loss and fragmentation. Similarly, the strong negative effect observed in Coleoptera supports the hypothesis that the UHI drives shifts to smaller body sizes in urban-tolerant ectotherms, consistent with Atkinson's temperature-size rule<sup>40</sup>. Larger beetles may also be more vulnerable to urban disturbance due to lower reproductive output and reduced dispersal capacity<sup>41</sup>. In contrast, urban communities of Lepidoptera (moths and butterflies) typically shift toward larger—and hence more mobile—species than rural communities<sup>15</sup>, likely because greater mobility facilitates persistence in typically fragmented urban landscapes<sup>42</sup>.

Order-level analyses (see Supplementary Text) offer further insight into the mechanisms shaping body size responses to urbanization. In plants, excluding grasses and ferns, we consistently observed a positive association between body size and urban tolerance. Taller species may gain a competitive edge in urban settings<sup>43</sup>, by exploiting higher temperatures, increased CO<sub>2</sub> and nitrogen availability<sup>44</sup>, and typically reduced O<sub>3</sub> concentrations<sup>45</sup>. In contrast, grasses and ferns favored smaller species, likely stemming from high disturbance regimes in urban grasslands and ruderal environments. Among ectotherm animals—including Diptera, Coleoptera, Hemiptera, and Squamata—the general expectation is a shift toward smaller species, consistent with metabolic constraints imposed by the UHI (Fig. 1 [↗](#), Fig. 3 [↗](#)). However, exceptions occur when urban fragmentation requires greater mobility, favoring larger and more mobile species<sup>15</sup>. Urban spiders also exhibit body size reductions, driven by both elevated temperatures and reduced prey size<sup>46</sup>. For endotherms such as rodents and birds, we typically anticipate a shift toward larger species because of increased habitat fragmentation, especially when body size correlates positively with mobility. While this pattern held for most bird families, notable exceptions—like Piciformes and Accipitridae—demonstrate negative associations, possibly due to constraints related to large home range requirements and prey size<sup>47,48</sup>. Future research should further explore such trophic consequences of urban size filtering, as these effects can either align across trophic levels or become mismatched. For instance, urban environments may favor larger predators—such as insectivorous birds and mammals—while simultaneously selecting for smaller insect prey species. Such mismatches can lead to nutritional stress, reduced reproductive success, and ultimately a more severe homogenization of urban predator guilds<sup>49,50</sup>. Another ecological consequence of size mismatches may occur when urban pollinator assemblages become disproportionally large or small relative to the floral traits of native plants, potentially reducing pollination efficiency<sup>51</sup>—although rapid evolutionary shifts in floral morphology could help mitigate these effects<sup>52</sup>.

## Current limitations and future directions

Our analysis, incorporating over 17,000 species, represents the most extensive assessment to date of how body size relates to urban tolerance across the tree of life. Such scale was enabled by the growing mobilization of biodiversity data through GBIF, particularly from citizen science platforms like iNaturalist<sup>53</sup>. Still, our analysis largely focused on common species, as we applied a cutoff of at least 100 observations per species per subrealm. As a result, rare or less frequently observed species are underrepresented, and their responses to urbanization remain an important





**Fig. 3. Effect sizes between body size and urban tolerance across the tree of life and individual effect sizes for animals and plants families.**

(A) Effect sizes of the relationship between body size and urban tolerance for the 371 families included in our analysis, plotted along a phylogenetic tree of life (see Methods); Plantae are highlighted and shaded in green. Colors indicate the direction of the effect: orange indicates negative, teal indicates positive, grey indicates neutral (i.e. any effect sizes between -0.05 and 0.05). (B) and (C) histograms of individual effect sizes for each family, for animals (B) and plants (C). Orders are shown along the outside edge of the phylogenetic tree, each with a bar and icon, for any order with more than 3 families. An interactive version for full exploration of our results at both family and order level is available [here](#).

avenue for future study. This is especially important given the growing recognition of cities as sanctuaries for various species<sup>37</sup>. Nevertheless, our findings highlight the relevance of common species to understanding macro-ecological patterns (*sensu*<sup>54</sup>). We focused here on broad-scale patterns, but future work should further explore the context-dependency of body size filtering—for instance, how it varies with city size, urban green infrastructure configuration, regional climate, and human cultural preferences. For example, one possible explanation for the observed trend of larger, urban-tolerant plants may lie in the widespread use of non-native woody ornamentals, such as alien trees and tall shrubs, which reflect human aesthetic preferences<sup>55,56</sup>—suggesting human preferences are an additional filter on plant size. In parallel, while our study examined interspecific variation, growing evidence also points to important intraspecific responses, where body size shifts within species can modulate urban tolerance<sup>31,57,58</sup>.

We anticipate that the urban tolerance scores presented here will form a valuable foundation for future research and local-scale planning efforts, in particular those that use citizen science to track restoration progress. For example, these tolerance scores could inform urban greenspace surveys aimed at calculating integrity indices—providing a repeatable, quantitative tool to assess and monitor the long-term success of urban ecological restoration by measuring the ‘urbanness’ of local species communities in urban greenspaces<sup>59</sup>. In this context, incorporating taxon-specific body size-urban affinity relationships can enhance conservation outcomes by tailoring strategies to the size dynamics of assemblages. For instance, in assemblages where larger species are disproportionately filtered out (i.e., a negative size-tolerance link), efforts should focus on increasing the size and quality of habitat patches and mitigating the UHI. Conversely, in assemblages where smaller, less-mobile species are more vulnerable, improving functional connectivity between habitat patches should be prioritized<sup>60</sup>. For butterflies in particular, Pla-Narbona *et al.*<sup>60</sup> advocate for such context-specific management strategies—balancing habitat patch quality and connectivity—to promote more diverse urban butterfly communities. Future research should also test the role of other key functional traits, like seed size (Fig. S2 [↗](#)), dispersal capacity (Fig. S3 [↗](#)), or thermoregulatory ability<sup>61</sup>, as analyses involving multiple functional traits will likely offer a better understanding of the mechanisms underlying species’ urban tolerance responses.

## Conclusions

Our global analysis of over 17,000 common species provides new insight into how diverse taxa respond to urbanization, revealing a consistent skew in urban tolerance—characterized by many urban avoiders and few exploiters—across biogeographic regions and taxonomic groups. We introduce the concept of Species Urbanness Distributions (SUDs) as a novel framework to quantify and compare the impact of urbanization on species communities. Much like Species Abundance Distributions (SADs) are essential for ecology and biodiversity research, SUDs offer a generalizable lens through which the community structure and ecological resilience of urban biotas can be assessed. Understanding the processes that generate SUDs, and understanding the ecological impacts of differently shaped SUDs, represent key avenues for future research. Although body size emerged as a predictor of urban tolerance, we found not only substantial heterogeneity across families and orders, but also that body size filtering alone is unlikely to explain the consistently skewed SUD shape. Nevertheless, trait-based approaches—especially those integrating multiple functional traits—hold strong potential for uncovering the processes driving the diversity of species’ urban responses and for interpreting the shape and skew of SUDs. Moreover, trait-based predictions of species’ vulnerability could be used to formulate effective strategies to promote biodiversity in urban environments<sup>62</sup>. Looking ahead, the continued growth of citizen science data platforms, such as iNaturalist, will play an increasingly critical role in tracking and understanding the mechanisms driving biodiversity responses to urbanization. These data streams will enable broader quantification of species’ urban tolerance, including rare and currently underrepresented taxa. Combined with trait-based modeling, this growing dataset could be leveraged to identify urban-vulnerable species and hence guide urban habitat restoration initiatives. In this way, SUDs and trait-informed predictions offer powerful tools for more effective urban strategies to conserve biodiversity on an increasingly urban planet.



## Methods

Our methodological approach can be broken down into three key steps: (1) quantifying species-specific urban tolerance scores, stratified by subrealm (i.e., biogeographic region); (2) collating measures of body size for as many species as possible for which we were able to quantify urban tolerance; and (3) quantifying the relationship between urban tolerance and body size at various taxonomic levels.

### Quantifying urban tolerance

Our goal was to derive a continuous metric of the extent to which species vary in their responses to urban areas (hereafter urban tolerance), ranking species from those that are largely intolerant (i.e., urban avoiders) to those that thrive in urban areas (i.e., urban exploiters). To quantify species-specific urban tolerances, we followed the approach described in detail in Callaghan et al.<sup>21,63,64</sup>.

Following Callaghan et al.<sup>63</sup>, we downloaded data from the Global Biodiversity Information Facility (hereafter GBIF) (<https://doi.org/10.15468/dl.4dcbgt>) on February 4th, 2021, including ~1.4 billion biodiversity records from >24,000 datasets. GBIF data were filtered to only include observations that were recorded as species, and an additional step was taken to ensure that the listed genus in GBIF matched the genus of the species name, ensuring validity of the taxonomic nomenclature of GBIF. Due to uncertainty in matching observations with remotely-sensed products, any GBIF observation with a coordinate uncertainty > 1 km was removed. We only included species from a taxonomic Class (i.e., the rank between Phylum and Order) that had at least 10 species reported to GBIF, focusing on the most common classes for downstream analyses. We then overlaid GBIF species occurrence observations with a remotely-sensed layer representing a continuous proxy of urbanization—Visible Infrared Imaging Radiometer Suite (VIIRS) night-time lights<sup>65</sup>. We used this proxy as it is a continuous measure of urbanization commonly used to represent urban extent<sup>66,67,68</sup>. Previous work has shown that VIIRS night-time lights is negatively correlated with greenness measured through the Enhanced Vegetation Index (EVI) and positively correlated with population density<sup>63,65</sup>. Further, previous work has also shown that measures of urban tolerance calculated from VIIRS night-time lights, population density, and global human modification index are all strongly correlated<sup>59</sup>. We used Google Earth Engine<sup>69</sup> for our geospatial processing and data were obtained for the VIIRS Stray Light Corrected Nighttime Day/Night Band Composites product, representing monthly composites, (i.e., this dataset in Google Earth Engine: *NOAA/VIIRS/DNB/MONTHLY\_V1/CMSELCFG* [link](#)).

We took the median of all monthly composites for each pixel to calculate a pixel-level urbanization value, measured in average radiance, and used imagery from January 2015 to January 2021 to calculate this median. GBIF data were filtered from January 1st, 2010 to February 4th, 2021. We acknowledge that these two temporal scales do not precisely match, but because urban conversion at the landscape/regional scale happens relatively slowly, we assume that the level of urbanization between 2015 and 2021 accurately captures the relative differences between regions with low and high urban land cover.

To avoid computational overload, we used geohash encoding to assign every record in GBIF to a “pixel” representing the VIIRS average radiance. We used geohash7 encoding to divide the geographic area into grid cells (referred to as blocks), each representing an approximate spatial area of 150 m<sup>2</sup>. Each GBIF record was assigned to a corresponding geohash7 block based on its geographic coordinates. The VIIRS night-time lights data, with a native resolution of ~500 m<sup>2</sup>, was then matched to these blocks by assigning each geohash7 block the average VIIRS radiance value that intersects it. While the geohash resolution is finer than the VIIRS data, this approach allows us to efficiently associate highly precise GBIF occurrence points with nearby VIIRS pixels using spatial indexing. This avoids unnecessary redundancy in the data while maintaining the original VIIRS resolution.

After each record from GBIF was assigned a measure of VIIRS, described above, for each species, we took the mean of the distribution of VIIRS values as a relative species-specific measure of a response to urban land cover (hereafter referred to as *urban score*). This urban score represents the relative ranking of that species to other species in response to urban land cover. To account for potential intraspecific differences in species' response to urbanization throughout their global range, we stratified our analysis to a total of 52 subrealms<sup>70</sup> (Fig. S1 [↗](#)). Therefore, a species could be represented across multiple subrealms with differing measures of urban tolerance (Fig. S4 [↗](#)). Importantly, this continuous metric of urban tolerance is a relative measure of a species' preference, or affinity, to urban areas: it should be interpreted only within each subrealm (i.e., urban tolerance scores are not interpretable across subrealms), providing a comparative or relative ranking of how a species responds to urbanization compared with other species within that subrealm. Within each subrealm, we further accounted for the potential of different levels of urbanization by scaling each species' urban score by subtracting the mean VIIRS of all observations in the subrealm (this value is hereafter referred to as *urban tolerance*). This 'urban tolerance' (Fig. S5 [↗](#)) value can be negative—when species under-occupy urban areas suggesting they actively avoid them—or positive—when species over-occupy urban areas suggesting they prefer them (i.e., ranging from urban avoiders to urban exploiters, respectively).

We applied the above procedure to any species that had at least 100 observations in a subrealm. Previous work has shown that this cutoff approximates the variability of a species' response to urbanization and ensures that enough of a species urban habitat use has been captured<sup>21,71</sup>. After this filtering we were left with a total of 56,181 data points—i.e., unique urban tolerance measures of a species in a subrealm—for potential inclusion in our analysis. These records represented 30,373 species that had at least one urban tolerance measure, from 47 classes and across 51 subrealms (Fig. S6 [↗](#); Data S1). The number of subrealms a species was found in was mostly (64%) 1 but ranged from 1 to 34 (Fig. S7 [↗](#)). All urban tolerance measures (i.e., urban tolerance by subrealm measures) which we produced as part of our workflow are made available, as we hope to advance further testing of urban tolerance among different taxonomic groups and regions. But correct interpretation of these urban tolerance values is essential before use.

Sampling biases are common in citizen science data. In GBIF, urban areas are better sampled compared to remote regions, as contributors typically concentrate around areas with high human density. However, sampling bias across the urban modification gradient was accounted for because each species' urban tolerance value was calculated relative to the observations across its geographic range (i.e., within the subrealm), compared with other species within the same geographic range. Importantly, while disproportionate sampling in human occupied areas may inflate the urban tolerance value, each species in the dataset was subject to similar biases, enabling comparisons across species. And hence, the focus here should be on comparisons among species. Previous work has found that this approach of assigning urban tolerance is strongly correlated with occupancy-detection models<sup>21</sup>. As illustrative examples, these data can be used to better understand the differences in urban tolerance among Hymenoptera in Northeast American Forests (Fig. S8 [↗](#)), Lepidoptera in Southeast Asian Forests (Fig. S9 [↗](#)), or Asterales in Scandinavia & West Boreal Forests (Fig. S10 [↗](#)).

## Species urbanness distributions

Our analysis of species urbanness distributions was descriptive in nature, relying on data visualization using ggplot2 to make histograms and density histograms of urban tolerance values for each subrealm. In these analyses at the subrealm level, species are treated equally without any differentiation taxonomically, compared with the taxonomically-stratified analyses where species within each taxonomic rank are treated as similar.

## Aggregating measures of body size

Our objective was to compile a dataset of species-specific measures of 'body size', or 'organism size' more broadly to integrate with our measures of species-specific urban tolerance described above. We aimed to find measures for as many species as possible, maximizing our sample size for

modeling. For plants, we used ‘plant height’ as a proxy for body size, and data were downloaded from the TRY database—a database compilation of plant traits from many different datasets<sup>72</sup>. For animals, body size can be measured in different ways, depending on the taxonomic group of interest (e.g., body length, wingspan, biomass, shell size, radius, forewing length, body mass, or biovolume). Therefore, our dataset was purposefully broad in its composition of body size estimates. Because ‘body size’ is the predominant form referred to in the literature, we use this throughout, but take it to broadly mean ‘organism size’ as discussed above.

To develop this dataset, we first performed a semi-systematic broad literature search, using Web of Science, Google Scholar, Zenodo, Dryad, and Figshare with variations of the key words: “body size”, “organism size”, “body length”, “body mass” and “interspecific”, “global patterns”, “animals”, and higher-level groups such as classes (e.g., “Aves”, “Birds”, “Mammals”, “Mammalia”). We started by downloading known compilations for various taxa, for example mammals<sup>73</sup> or Odonata<sup>74</sup>. In these instances, we spent minimal further time searching for specific body size data for these larger taxonomic groups, assuming that these compilations had already aggregated the majority of data available. Following this, we conducted more detailed literature searches, predominantly using Google Scholar and Web of Science, which focused on our *a priori* list of species for which we had a measure of urban tolerance (Data S1). These were conducted at class, order, or family taxonomic levels, but not at taxonomic levels below family. Multiple keywords were used for each group while searching when it was obvious to do so. For example, when searching for body size measures in Coccinellidae both “ladybirds”, “ladybugs”, and “Coccinellidae” were used in search strings, as well as “wing length”, “body length”, or “body size”. Search effort was qualitatively proportional to the number of species we had in each taxonomic level. For example, because we only had 43 species with urban scores from the class Diplopoda but 3,515 species from the order Lepidoptera, we spent proportionately more time and effort searching for body size data for Lepidoptera. Similarly, we spent more time on hard-to-find taxa, namely groups from Insecta. In our searches we generally ignored studies that were focused on intraspecific variation in body size for one or a few species, unless we found that the species was included in our urban tolerance list. In addition to literature searches, we emailed networks of colleagues, looked at reference lists in papers that had data, and emailed corresponding authors of papers who had potentially relevant data but for which data were not accessible.

After this semi-systematic searching, we re-assessed the remaining species for which we had an urban tolerance measure but not a measure of body size. We then performed a second-level targeted search (i.e., more systematic) for any family with at least 10 species remaining for which we had a measure of urban tolerance. First, we searched Web of Science and Google Scholar one family at a time and looked at the first 100 hits, with the family name ‘and’ “body size” used as search terms. Second, we used Google search engine to search for these remaining species individually (N~3,100). We focused on including species for which Google immediately returned an estimate of body size (e.g., through Wikipedia, or an aggregating website), and did not go to original species descriptions. This detailed searching, and the aggregation of body size measures, concluded in June 2022.

## Measurements of body size

In our searching for body size, we were agnostic to the type of body size, given that many different measures are used for quantifying body size, or proxies for body size. For each dataset we downloaded or processed, we kept track of the following key metadata properties: ‘type’, ‘units’, and ‘measurement detail’. Type refers to the different types of body size measurement, ranging from body length and body mass to Weber’s length in ants. Table S1<sup>75</sup> shows a list of the different body size types encompassed in our analysis (see statistical details below for how different body size types were accounted for in our analyses). For each dataset, we also kept track of the units (e.g., mm, g, mg cm<sup>-2</sup>, or mm<sup>3</sup>). And lastly, wherever possible, we noted the measurement detail of the body size measures, whether the mean, median, or maximum was used, or whether it was for males or females only. Many times, however, this information was not readily available, and thus this metadata field was not required in our dataset composition.

## Taxonomic harmonization and dataset integration

Because body size data were aggregated across many potentially different datasets, we performed a taxonomic harmonization step to regain some species that did not match due to typos, spelling mistakes, or synonyms. For this, we treated Plantae and Animalia separately and used the `taxize` package<sup>75</sup> in R to search through the list of species and find any synonyms or fuzzy matches of other taxonomic entities, always matching back to the GBIF taxonomic backbone, given that our urban tolerance measures follow this taxonomy (Data S1). With the possible synonyms and matches, we then integrated the urban tolerance measures with the body size dataset using either direct matches or matches of synonyms. The taxonomic harmonization step regained a total of 831 (see Fig. S11 [↗](#)) species that were integrated in the dataset for analysis. Data S3 provides details on the 223 different ‘datasets’ (i.e., downloaded data, or manually added data) that were processed as measures of body size. The most prevalent class in our final dataset (Fig. S12 [↗](#)) was Insecta (N=7,526), followed by Magnoliopsida (N=4,872) and Aves (N=3,313).

Our final dataset for potential analysis and modeling (Data S2) included a total of 94,087 observations (i.e. unique combination of species urban tolerance values, subrealm, and body size measure) for 20,957 species that had at least one measure of urban tolerance and at least one measure of body size.

## Quantifying the relationship between urban tolerance and body size

Our objective here was to understand the relationship between urban tolerance (the response variable) and body size (the predictor variable). Our dataset had potentially multiple measures of body size for an individual species, and differing levels of metadata. For example, some datasets may have specified whether the mean was used, or others might have specified for females only (see Table S1 [↗](#)). Therefore, before modeling, our goal was to minimize the undue effect of having too many random effect levels. First, because an individual species could have multiple measures of body size, we only selected the body size measure that was most common among the species in each family. For example, if a species had both a measure of body length and of body mass, we selected the measurement type that was most commonly available among the species in each family and the largest source of data. Second, we further reduced the potential number of levels and variation in body size measurement by creating a synthetic “metadata” variable that was the combination of ‘type’, ‘units’, and ‘measurement detail’. By doing this, we decreased the potential number of levels of a random effect from 223 (i.e., the number of data sources) to 71 (i.e., the combination of type, unit, and measurement detail). Third, several papers/data sources could have all reported the same metadata (e.g., maximum body length for females in mm), and instead of being treated as individual random effects because they were from different data sources, we assumed that these represented similar data and collapsed them into one random effect level (here, for maximum body length for females in mm).

## Overall kingdom-level model construction

We used Bayesian hierarchical models to test for the effect of body size on urban tolerance. In our models, urban tolerance was the response variable and body size was the predictor variable of interest. To evaluate the overall relationship between body size and urban tolerance, we fitted two comprehensive models that incorporated hierarchical taxonomic levels and the relationship with body size. The first model included fixed effects for kingdom and log-transformed body size due to its skewed distribution, along with their interaction, to investigate differences in the body size effect on urban tolerance between plants and animals. The random effects accounted for hierarchical taxonomic levels, incorporating random intercepts and slopes for class, order, and family to capture variation at each level. Additional random effects for metadata and subrealm were included as random intercepts to control for study-specific differences in data and regional differences in urban tolerance. The model formula was:  $\text{urban tolerance} \sim \text{body\_size\_scaled\_log10} * \text{kingdom} + (1 + \text{body\_size\_scaled\_log10} \mid \text{class/order/family}) + (1 + \text{body\_size\_scaled\_log10} \mid$

metadata) + (1 | subrealm). The second model built upon this structure but only included an interaction term between log-transformed body size and kingdom as well as kingdom as a fixed effect to determine if the effect of body size on urban tolerance varied consistently across the plant and animal kingdom. It maintained the same hierarchical random effects structure as the first model, with class, order, and family nested to account for hierarchical variation, and metadata and subrealm included as additional grouping factors. The formula was:  $\text{urban tolerance} \sim \text{kingdom} + \text{body\_size\_scaled\_log10}:\text{kingdom} + (1 + \text{body\_size\_scaled\_log10} | \text{class/order/family}) + (1 + \text{body\_size\_scaled\_log10} | \text{metadata}) + (1 | \text{subrealm})$ .

## Family-specific model construction

However, for our main analyses we chose to employ a ‘many models’ approach, where we fit individual models for each family that met a set of minimum criteria because many families do not have a known association or published information on the extent to which urban tolerance associates with body size. A family was only included if there were at least 10 unique species with an urban tolerance measure and body size measure, and a subrealm was only included if it had at least two species included to specify a random slope. For example, if there was only one species of a particular family included in our dataset, this family would not have received an independent model. We decided to focus our main analysis at the family level for three key reasons. First, this was done because for most of the taxonomic families included in our analyses there is little *a priori* expectation about the effect of body size on urban tolerance (i.e., little to no literature on the subject), and therefore we wanted each model to represent an advance in knowledge of that relationship for that family. Second, because there were large imbalances in the number of data points available for each taxonomic group, for instance some families having hundreds of species with available data and others with only a handful of species, we wanted to avoid shrinkage where the model results were primarily driven by the taxonomic groups with a larger number of data points represented. Third, we wanted to avoid an instance where a family (or other taxonomic grouping) could have little or no relationship between body size and urban tolerance, but if included together in one model could result in a positive effect (see Fig. S13 for a simulated example). This was a smaller concern at the family level, which we present as the main analysis, because of the smaller imbalance in data availability across the taxa within each family.

When we fit the family-specific models, there were different requirements for the random effect structure depending on how many subrealms and metadata sources (i.e., types of body size measures). In the simplest case, all species in a family were just from one subrealm and had just one data source for body size. But this was not always the case, and therefore we developed four different ‘types of models’ which varied in their treatment of random effect structure. Each ‘type’ of model, however, had the same overall objective—to quantify the relationship between body size and urban tolerance. The four model types are as follows. Type 1: had at least two subrealms and at least two types of synthetic metadata (see above) sources included. Type 2: had data from only one subrealm and only one type of synthetic metadata source included (but could have data from more than one initial data source). Type 3: had at least two subrealms and only one type of synthetic metadata source included. Type 4: had data from only one subrealm and at least two types of synthetic metadata sources included. For all types of model fits, the response variable was the urban tolerance measure, described above, and the fixed effect of interest was body size which was log10-transformed and scaled and centered by the synthetic metadata source. Type 1 had a random intercept and slope for subrealm, allowing the intercept and slope to vary across different levels of subrealm as well as a random intercept for the synthetic metadata source. Type 2 had no random effects. Type 3 had only a random intercept and slope for subrealm, allowing the intercept and slope to vary across different levels of subrealm. Type 4 had only a random intercept for the synthetic metadata source. Modeling was done using the ‘`brm()`’ function in R from the brms package<sup>76,77,78</sup>. For all model types we specified a standard normal prior, with a mean of 0 and a standard deviation of 1 applied to the fixed effects of the model. Models were fit with 1000 warmup iterations, 6000 iterations for the Markov Chain Monte Carlo sampling, 4 chains, and the adaptation target acceptance probability was set to 0.99.



For models of type 1 and type 4, where we combined metadata for body size measurements among different species, we also explored more complex model structures that included a random slope for body size by metadata source, in addition to subrealm. Such a model structure would allow the relationship between body size and urban tolerance to vary across measurement types. However, these models often substantially reduced interpretability and inflated uncertainty around slope estimates, especially when metadata groups were sparsely represented, which was common. Additionally, many of the families only had 2 types of metadata, limiting our statistical power to fit random slopes. Comparative analyses showed that such models produced weaker or more uncertain estimates than our main model structure (i.e., random intercept for metadata of body size). We also note that to account for heterogeneity in body size measurement types, all body size values were log10-transformed, scaled, and centered by their metadata source prior to analysis, following standard practices in allometric studies. We use two illustrative examples (Figures S14 [↗](#) and S15 [↗](#)) to illustrate the influence of including a random slope for metadata source.

## Additional analyses beyond family level

Although our main analysis was focused on the family level, one potential drawback is that species could be excluded from any of the analyses for a family that has a small number of species. Plus, because some analyses have previously been conducted at higher taxonomic levels (e.g., Lepidoptera), we repeated the above procedure described for family at the order (see Supplementary Text) and kingdom level.

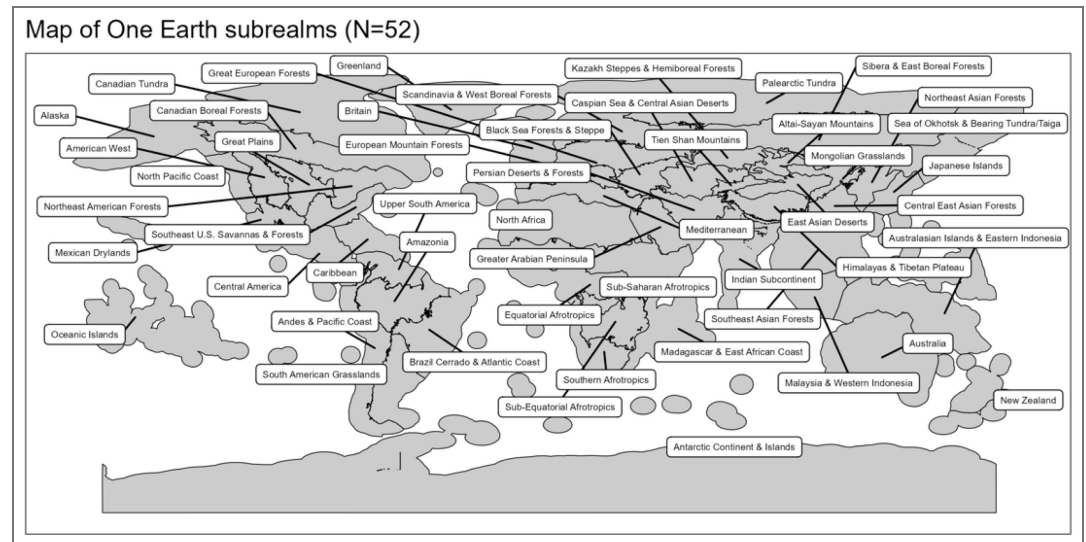
## Statement of authorship

CTC, DEB, TM conceptualized the project and developed project methodology. CTC, TM, VS, JHW, BMM conducted analyses. CTC conducted project administration and supervision. CTC, DEB, TM wrote the original draft, and CTC, DEB, VS, BMM, LA, IS, JHW, TM were involved in review and editing.

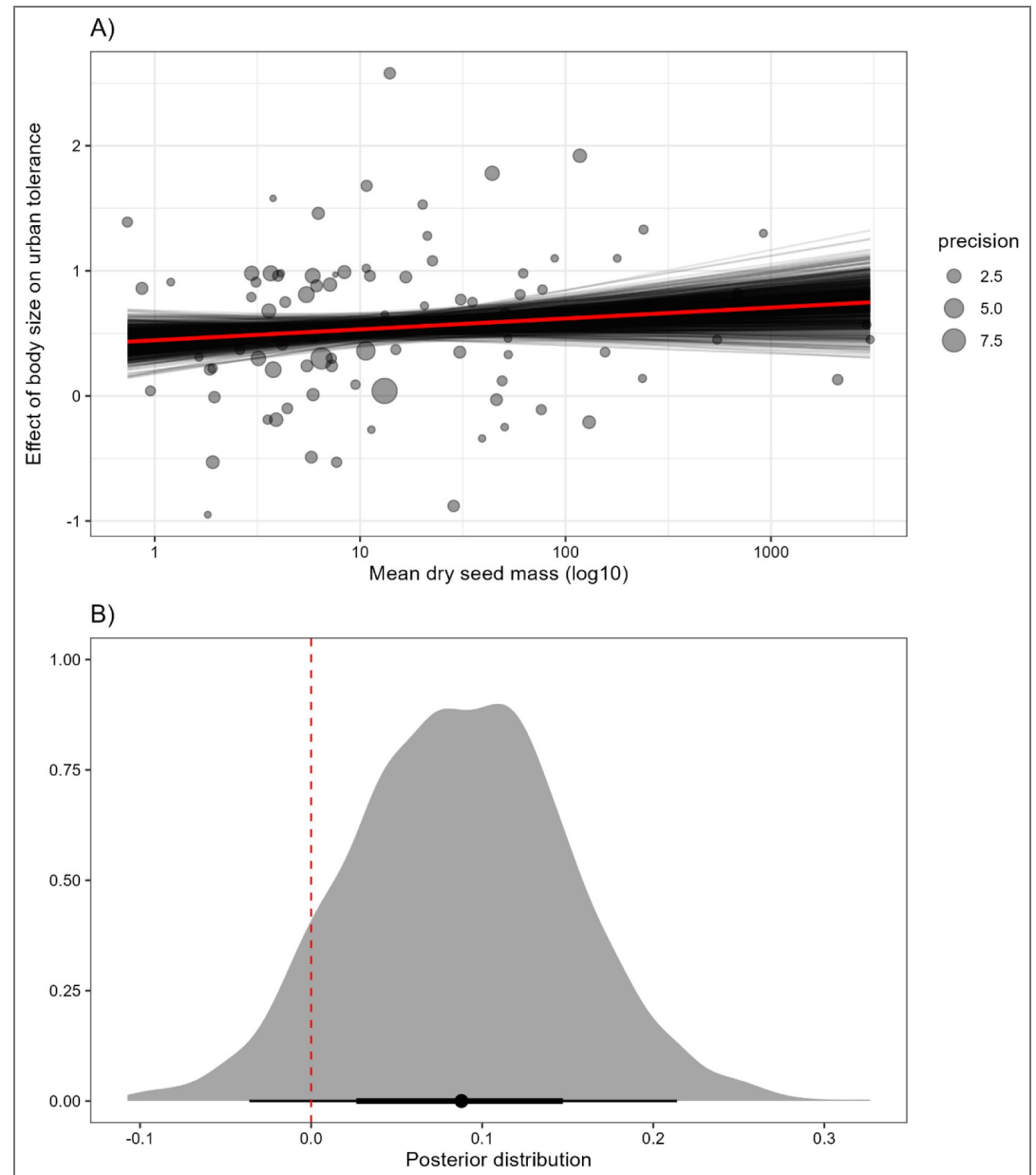
## Supplementary Text

In addition to family-level effect sizes, we performed the analysis at the order level, and found similar heterogeneous results. Among the 128 orders, a total of 45 had some evidence of an effect of body size on urban tolerance (14 weak, 6 moderate, and 25 strong). Some of the orders with the strongest positive effect size included Sapindales (dicots, soapberry plants) and Columbiformes (birds, doves and pigeons), and orders with the strongest negative effect sizes included Coleoptera (insects, beetles) and Piciformes (birds, woodpeckers). Of note are the differential responses down the taxonomic guilds, where some families can show contrasting results, but there may be a strong effect when looking at the order level, or higher (e.g., Fig. S2 [↗](#)).

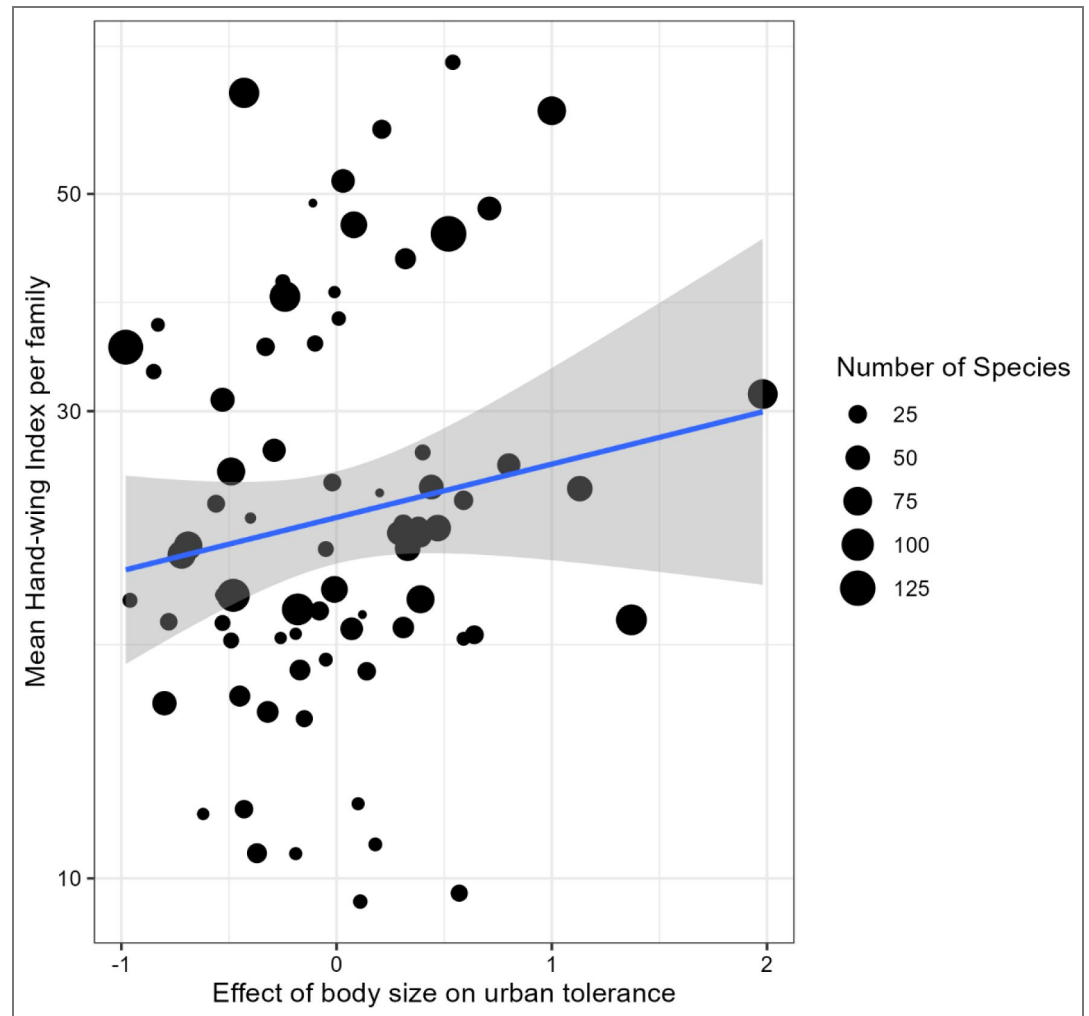




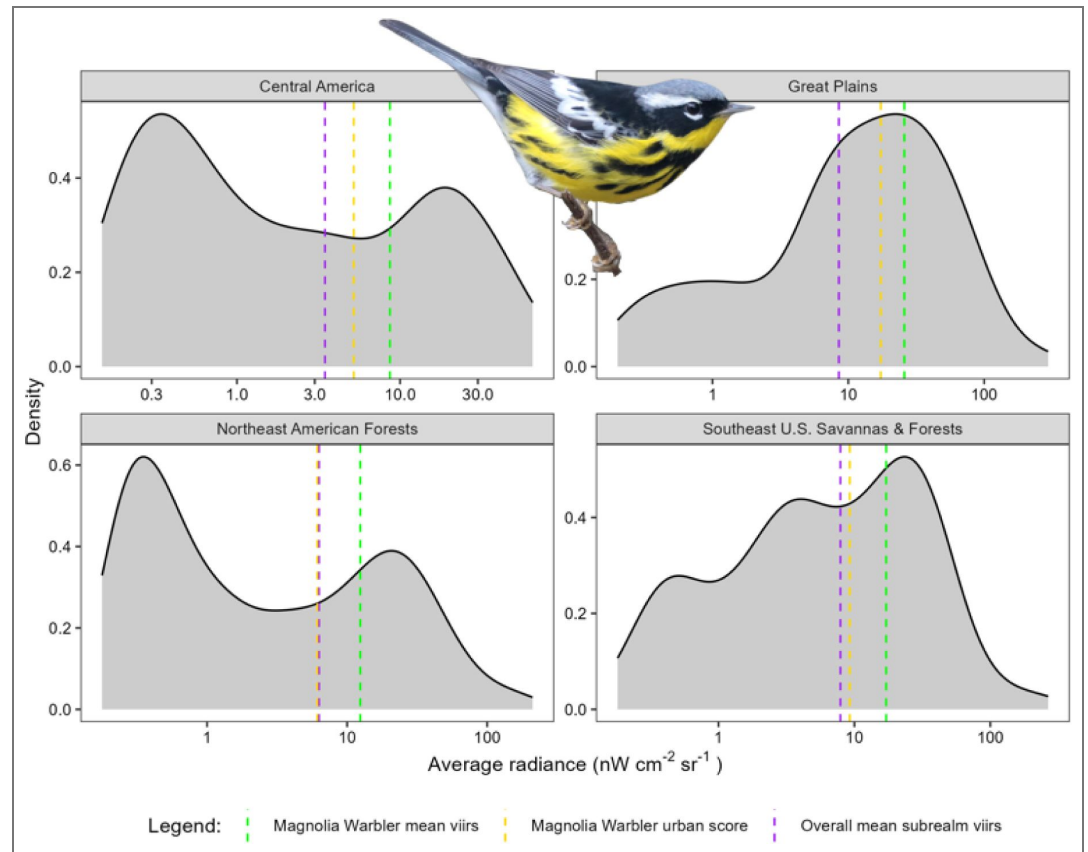
**Fig. S1. We used subrealms as our geographical aggregation.** Subrealms were quantified from aggregating bioregions as identified by One Earth (see more here: <https://www.oneearth.org/bioregions/>). The subrealms level was chosen after exploring the tradeoff between accounting for geographic differences in urban tolerance and the number of species that could be included.



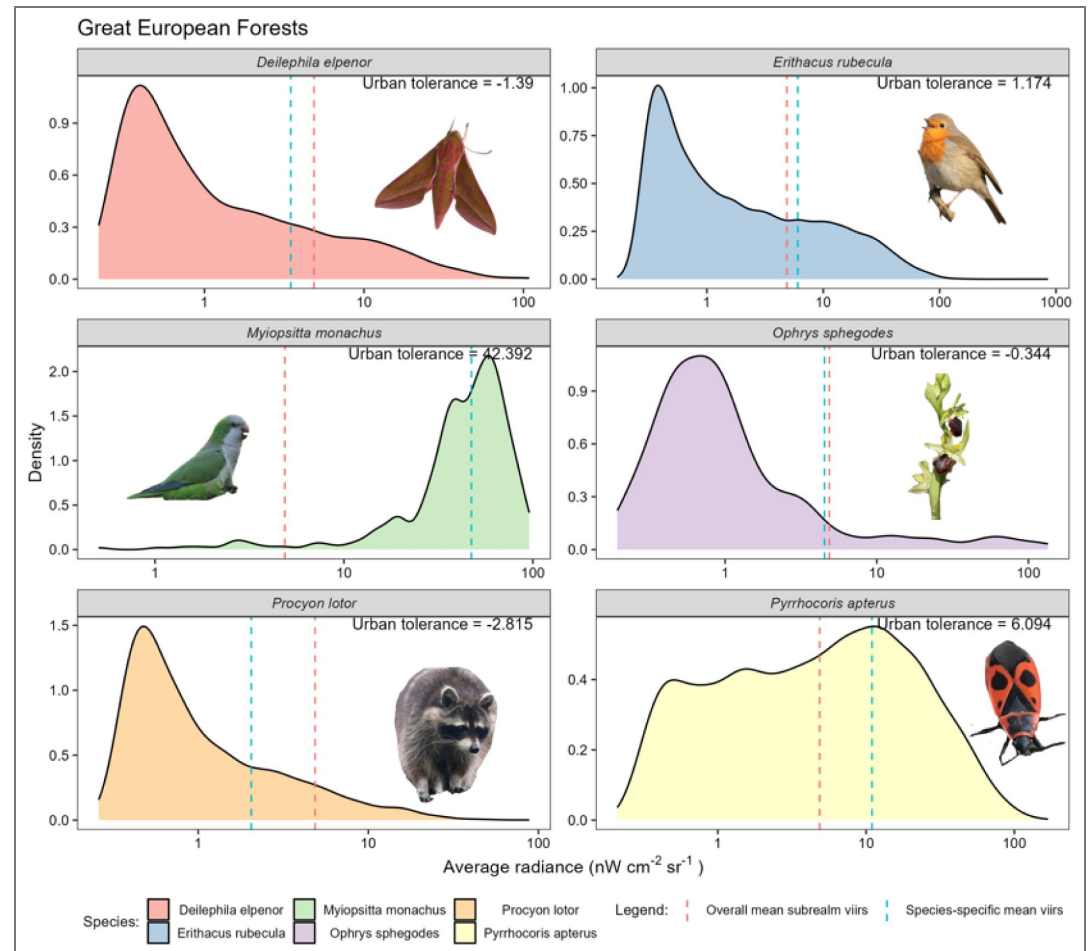
**Fig. S2.** (A) Exploratory analyses show some correlation between the effect of body size on urban tolerance for plants and the mean dry seed mass of those plants. (B) A Bayesian model finds moderate evidence that this is a positive relationship. This indicates that dispersal may be an underlying mechanism influencing the effect of body size (i.e., plant height) on urban tolerance.



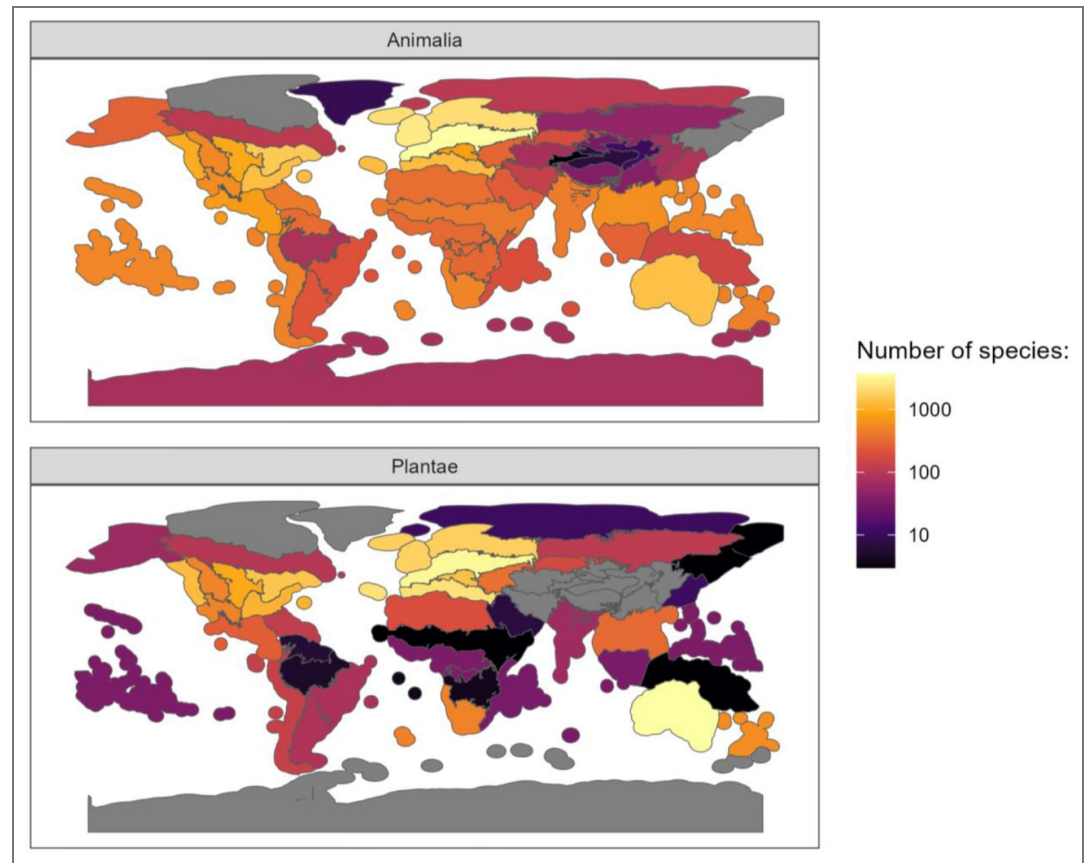
**Fig. S3.** Exploratory analysis for 74 bird families, showing a potential relationship between dispersal (measured using the mean hand-wing index per family) and the effect of body size on urban tolerance. This indicates that dispersal may be an underlying mechanism influencing the effect of body size on urban tolerance. The blue line is the linear model trend line and the grey shading represents the 95% confidence interval.



**Fig. S4.** The distribution of Magnolia warbler *Setophaga magnolia* observations and the VIIRS values, in average radiance, of those observations in four different subrealms. Shown using dashed lines are the species-specific mean (green), the subrealm-specific mean of VIIRS (violet), and the resulting urban tolerance measure (yellow).

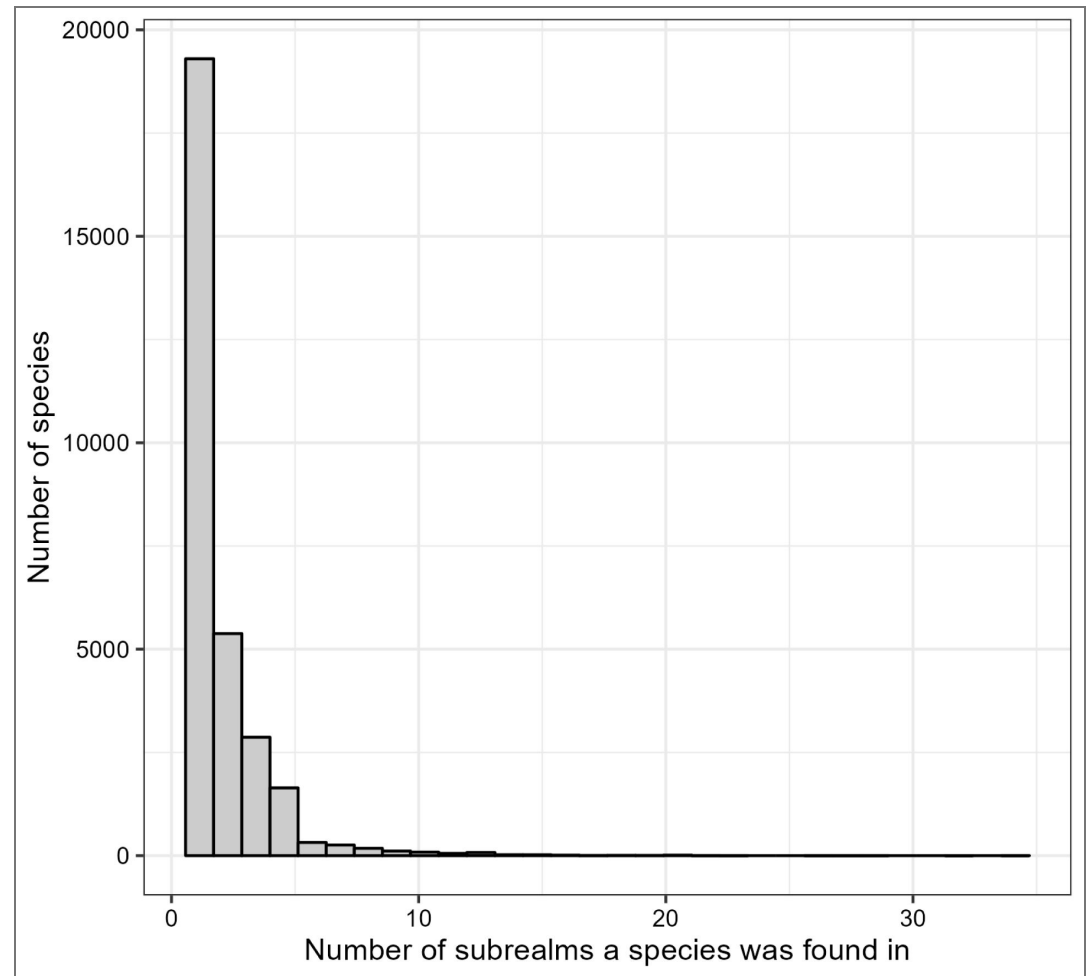


**Fig. S5.** An illustration of the species-specific distribution of observations and the VIIRS values, in average radiance, of those observations for six different species within the Great European Forests subrealm. Of note is that the urban tolerance is not shown, as some are negative values, but these values are shown in the text.

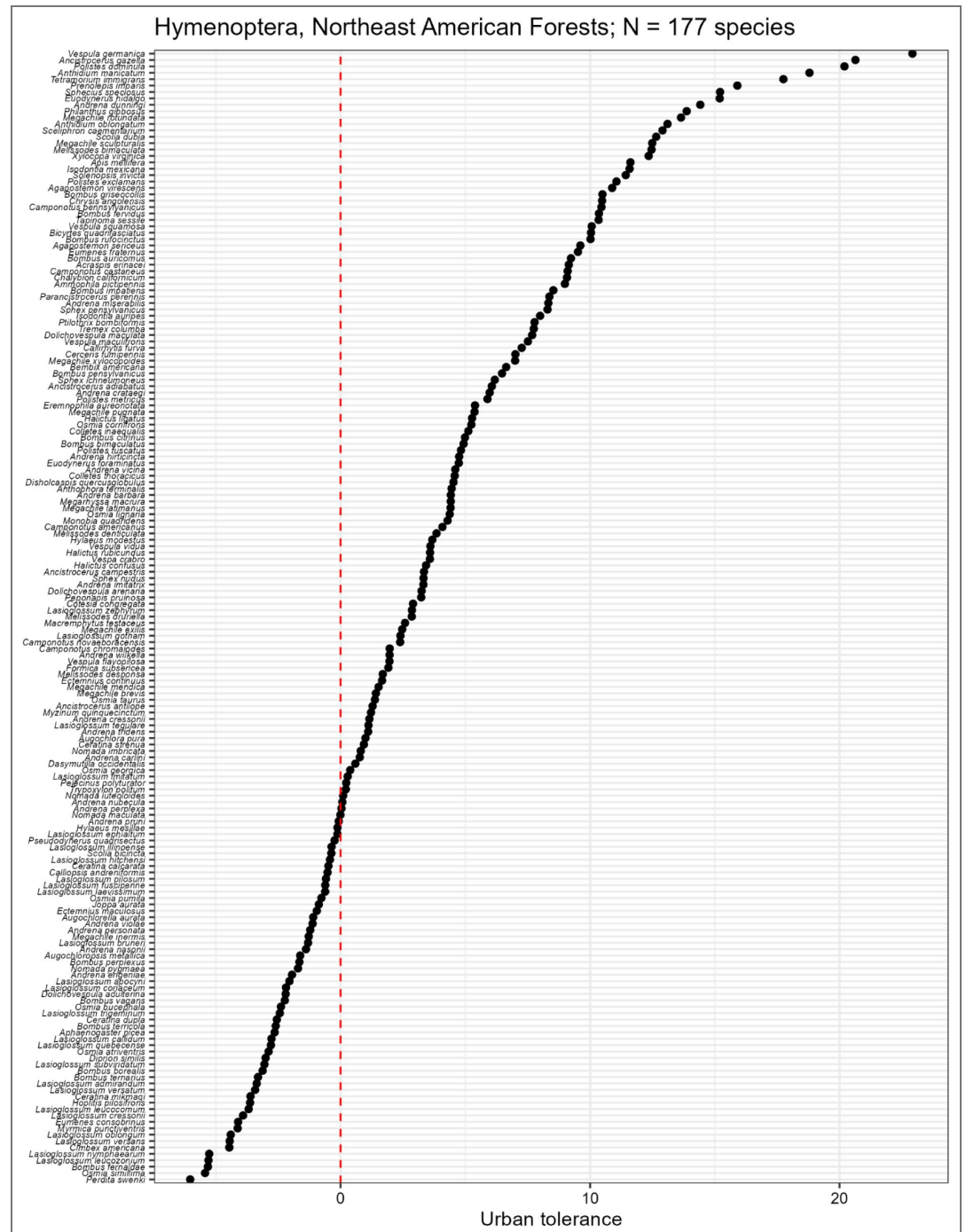


**Fig. S6.** The total number of species for which we calculated urban tolerance scores, stratified by subrealm and shown separately for Animalia (top) and Plantae (bottom). The subrealms with their names are shown in Figure S1. The dataset of species' urban tolerance scores is provided in Table S1.

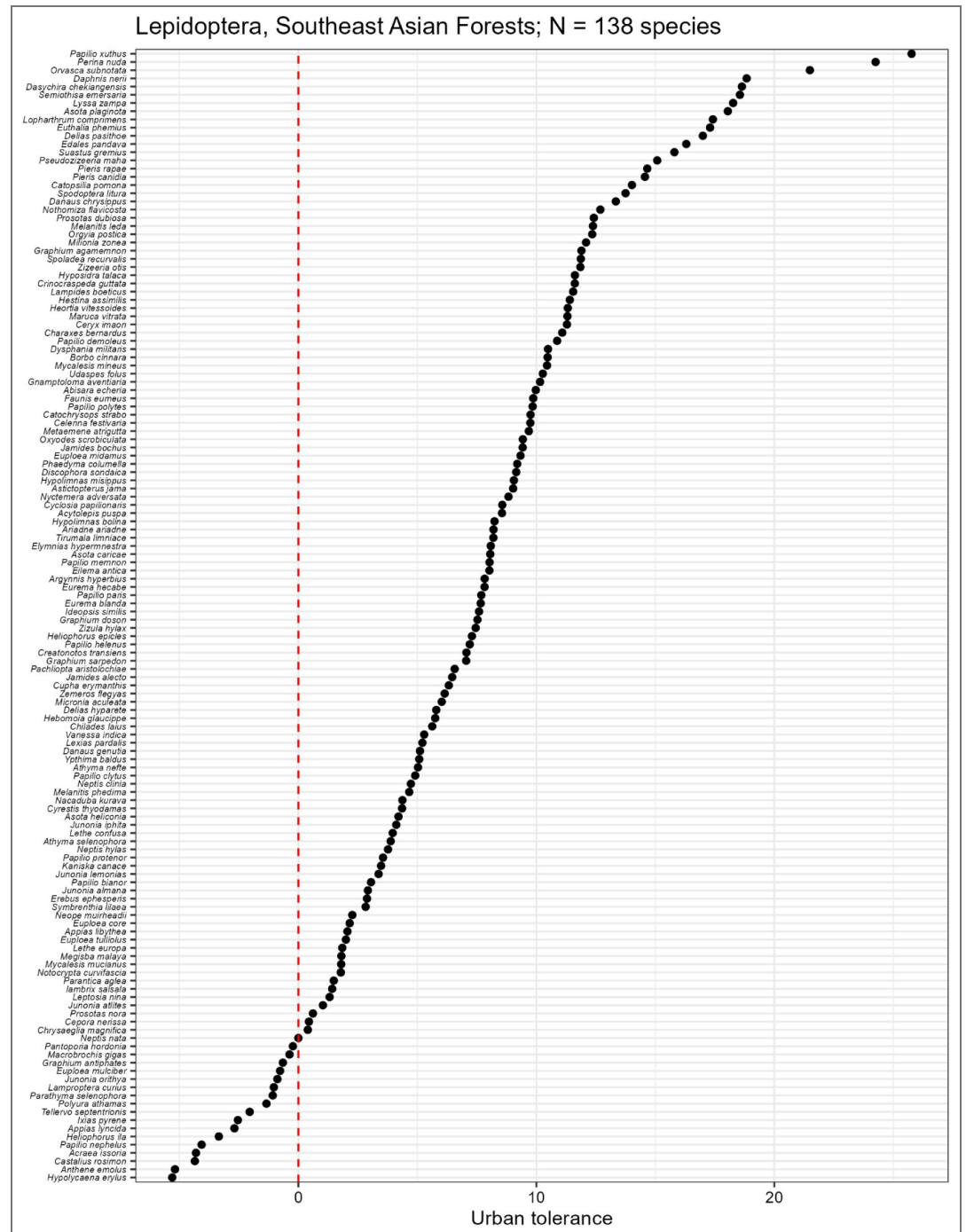




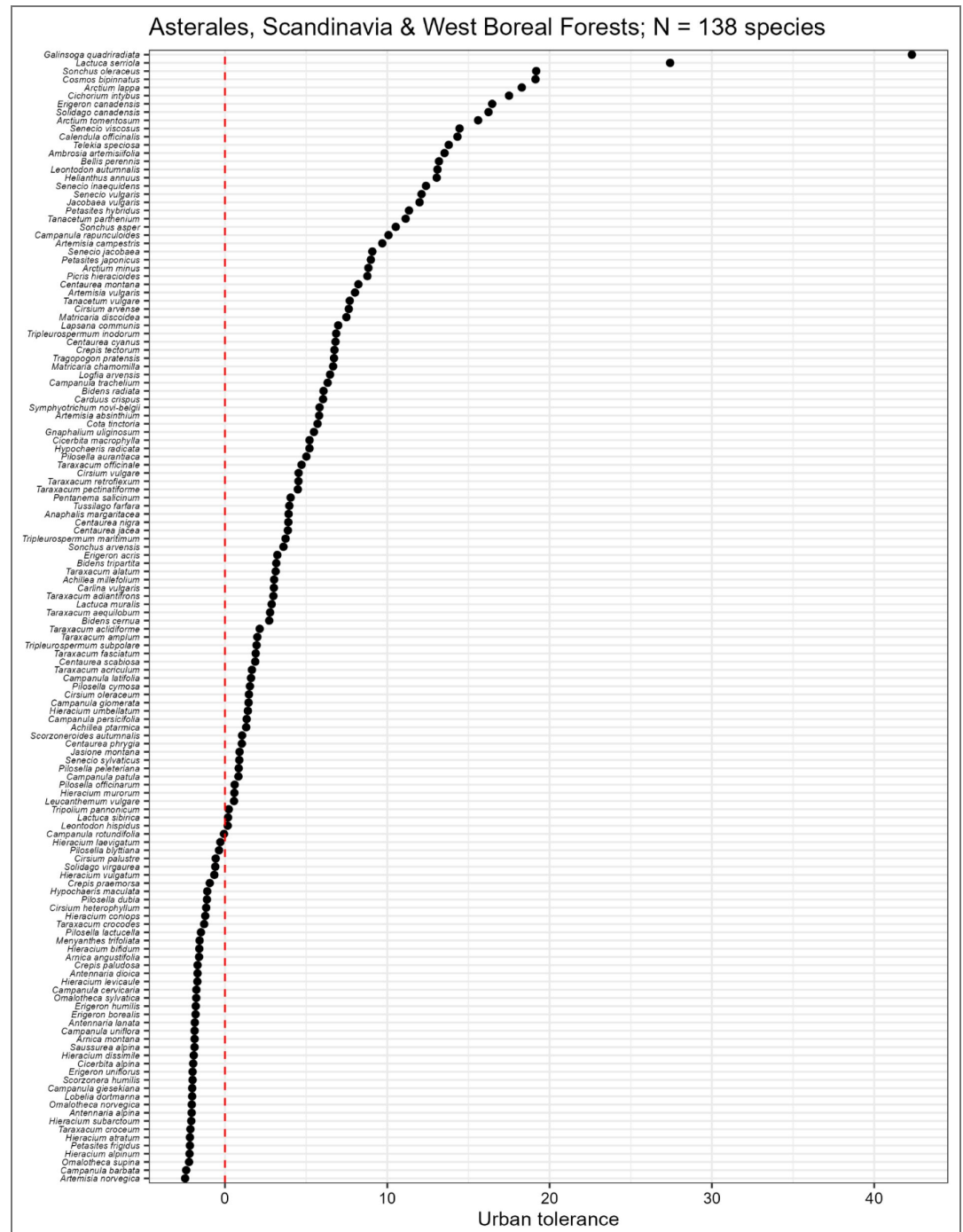
**Fig. S7.** The total number of subrealms for which a species had an urban tolerance score. The majority of species (64%) only had an urban tolerance score from one subrealm, but the range was from 1 to 34.



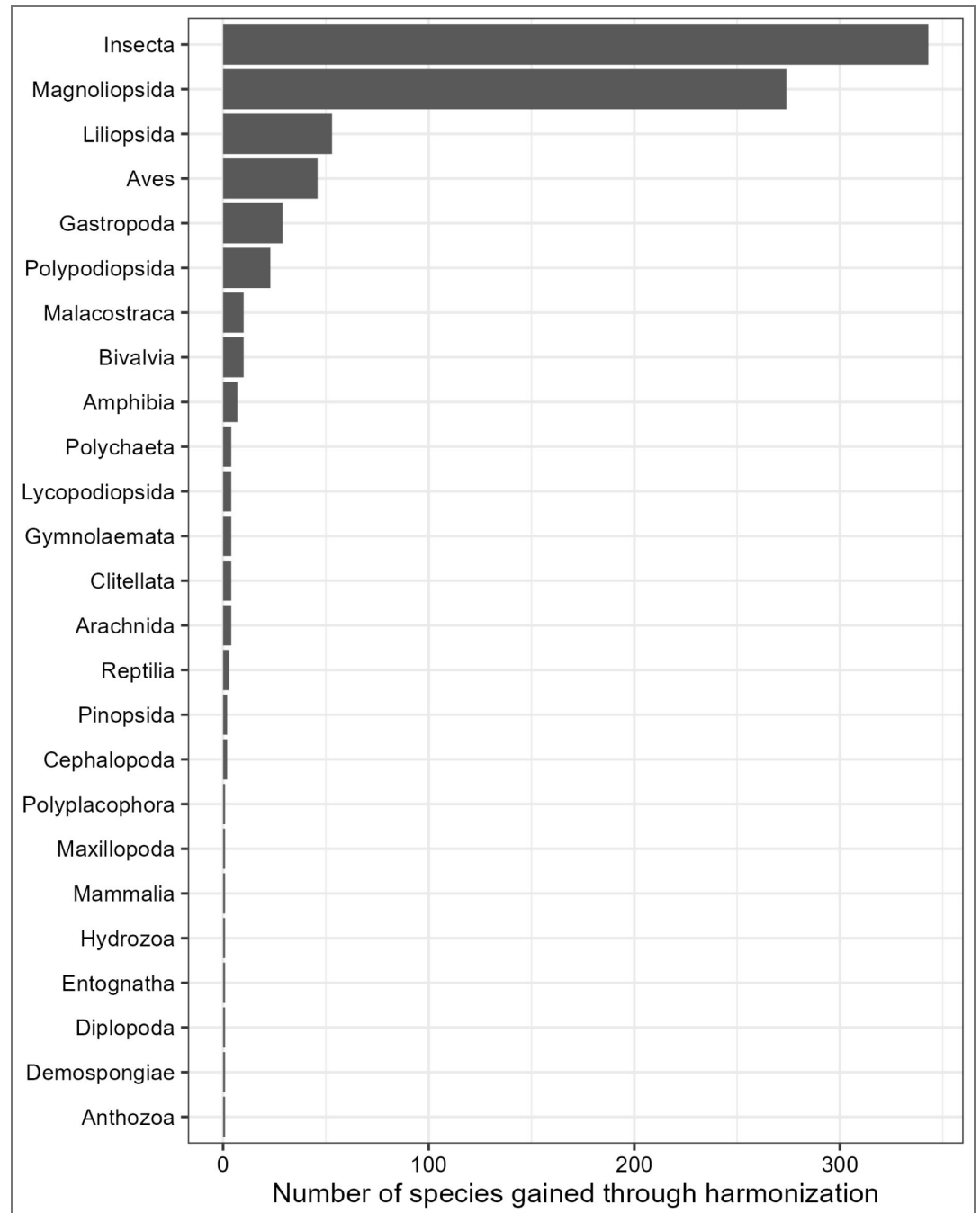
**Fig. S8.** An illustrative example, showing the relative urban tolerance scores for Hymenoptera in the Northeast American Forests subrealm. The plot is for illustrative purposes and the values for each species can be found in Data S1.



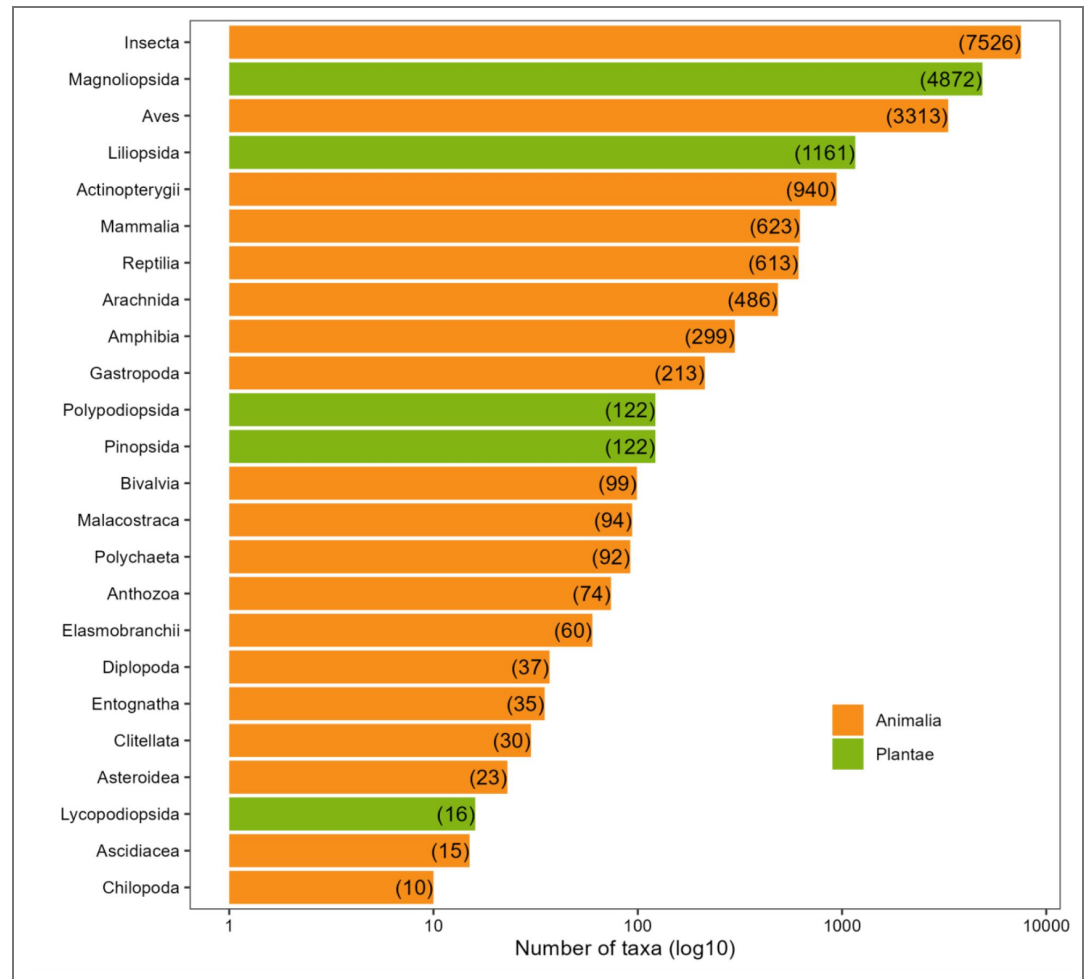
**Fig. S9.** An illustrative example, showing the relative urban tolerance scores for Lepidoptera in the Southeast Asian Forests subrealm. The plot is for illustrative purposes and the values for each species can be found in Data S1.



**Fig. S10.** An illustrative example, showing the relative urban tolerance scores for Asterales in the Scandinavia & West Boreal Forests subrealm. The plot is for illustrative purposes and the values for each species can be found in Data S1.

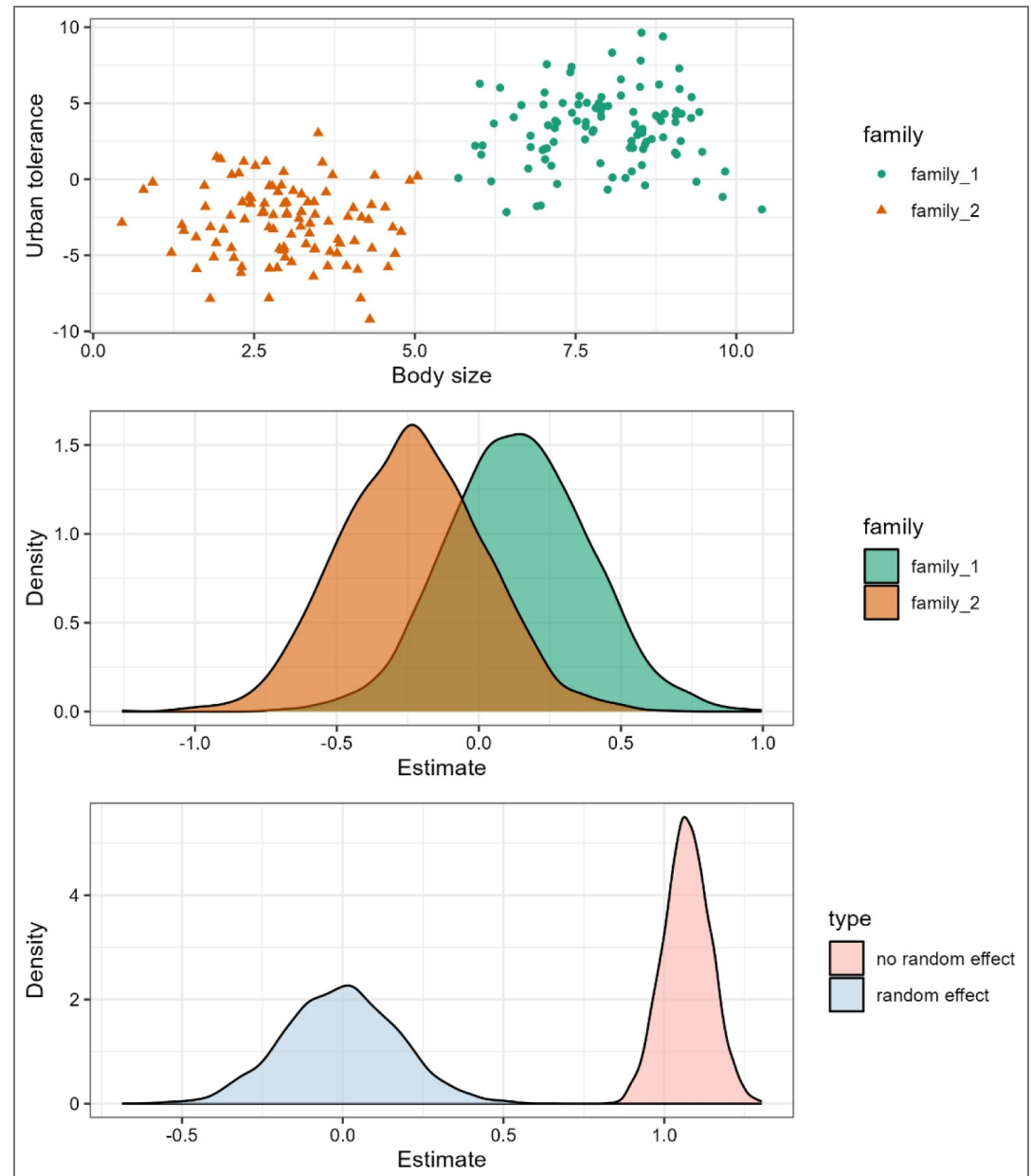


**Fig. S11.** A summary of the number of species, shown per class, gained through taxonomic harmonization, and therefore included in the analysis dataset.

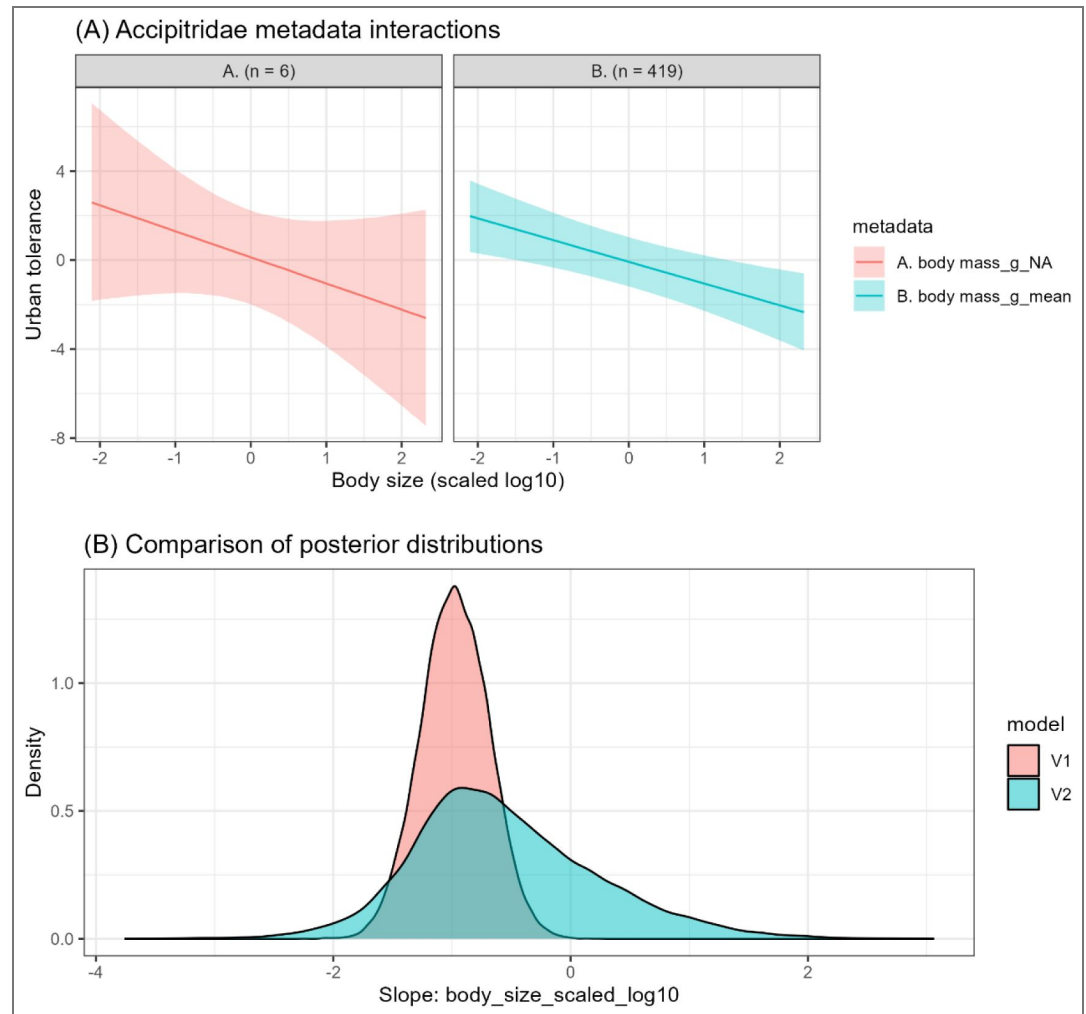


**Fig. S12.** The number of taxa, on a log10-transformed scale. These can be found in Table S3.

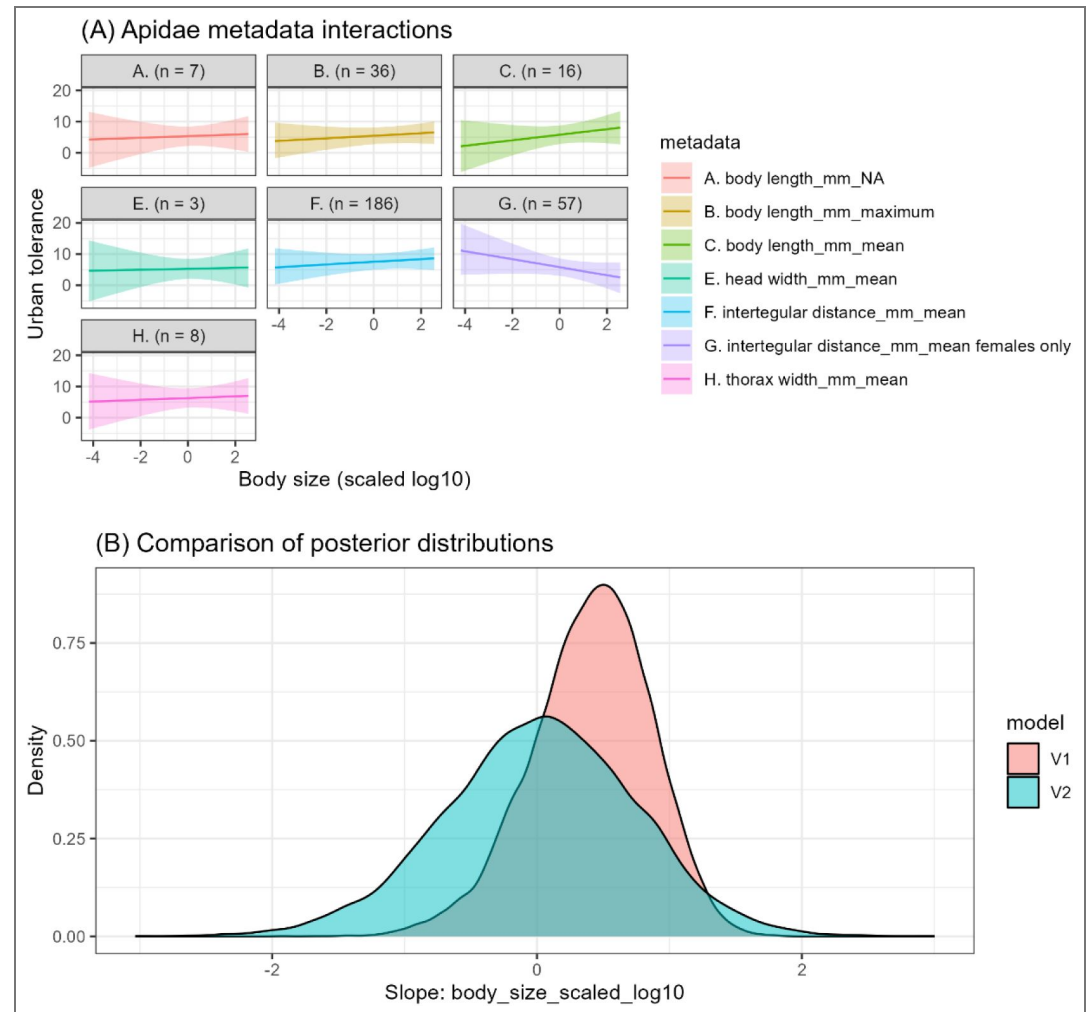




**Fig. S13.** We chose to model each taxonomic group (i.e., family, order, class, phylum, and kingdom) independently of one another to avoid the influence of where individual families could have no effect but result in a positive effect if modeled jointly. The top panel shows simulated raw data for two families; the middle panel shows the posterior distribution of a bayesian model fit for each of these families separately; the bottom panel shows the posterior distribution of a bayesian model fit for a model including family as a random effect and for a model not including family as a random effect.



**Fig. S14.** An illustrative example showing the influence of using a random slope for body size, by showing the (A) Interaction between body size (scaled and log10 transformed) and body size measurement type (metadata; see Methods for details) for the family Accipitridae. Facets represent different types of body size metrics used in the dataset, labeled with a letter identifier and corresponding sample size (n). Lines represent predicted urban tolerance based on body size, with 95% credible intervals shown as shaded ribbons. These are extracted from a model fit with an interaction between metadata and body size to purposefully investigate the influence of metadata and the relationship with body size. (B) Posterior distributions of the slope for body size (scaled and log10 transformed) under two model structures: V1 includes a random slope for body size by subrealm and a random intercept for metadata (as presented in main results), while V2 adds a random slope for body size by metadata. The inclusion of this additional random slope in V2 increases uncertainty and pulls slope estimates toward zero, particularly when data are sparse within body size measurement types as illustrated here. Also see Fig. S15.



**Fig. S15.** An illustrative example showing the influence of using a random slope for body size, by showing the (A) Interaction between body size (scaled and log10 transformed) and body size measurement type (metadata; see Methods for details) for the family Apidae. Facets represent different types of body size metrics used in the dataset, labeled with a letter identifier and corresponding sample size (n). Lines represent predicted urban tolerance based on body size, with 95% credible intervals shown as shaded ribbons. These are extracted from a model fit with an interaction between metadata and body size to purposefully investigate the influence of metadata and the relationship with body size. (B) Posterior distributions of the slope for body size (scaled and log10 transformed) under two model structures: V1 includes a random slope for body size by subrealm and a random intercept for metadata (as presented in main results), while V2 adds a random slope for body size by metadata. The inclusion of this additional random slope in V2 increases uncertainty and pulls slope estimates toward zero, particularly when data are sparse within body size measurement types as illustrated here. Also see Fig. S14.

| Type of body size     |                       |                   |                     |
|-----------------------|-----------------------|-------------------|---------------------|
| Weber's length        | abdominal length      | biovolume         | body length         |
| body mass             | body mass female only | carapace width    | cephalothorax width |
| colony height         | column diameter       | coral diameter    | diameter            |
| dorsal mantle length  | dry mass              | elytron length    | femur length        |
| forewing length       | fork length           | head height       | head length         |
| head width            | height                | hind tibia length | hindwing length     |
| intertegular distance | longest length        | plant height      | pronotal width      |
| radius                | shell size            | shell volume      | snout vent length   |
| standard length       | thorax length         | thorax width      | total biomass       |
| total length          | wet mass              | width             | wing length         |
| wingspan              |                       |                   |                     |

**Table S1.** A list of 41 potential 'types' of body size that were used for potential inclusion in our body size dataset. We aimed to incorporate as many types of body size measures as possible and were not restrictive in our searching for body size measures.

## Data accessibility

All data analysis was conducted in R statistical software and relied heavily on tidyverse (Wickham et al. 2019). The doi representing our download is: <https://doi.org/10.15468/dl.4dcbgt>. Our processing of the GBIF data to quantify urban scores is available here: [https://github.com/coreytcallaghan/body\\_size\\_vs\\_urban\\_tolerance](https://github.com/coreytcallaghan/body_size_vs_urban_tolerance). Not all body size data can be openly shared due to copyright restrictions, however the freely available dataset, cleaned and processed for our analysis is available here: [https://github.com/coreytcallaghan/body\\_size\\_vs\\_urban\\_tolerance](https://github.com/coreytcallaghan/body_size_vs_urban_tolerance). In this repository we did provide reproducible data and code, albeit not for the entire dataset of body sizes. We will archive a cleaned version of this repository in Zenodo upon acceptance of this manuscript.

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## Peer reviews

### Reviewer #1 (Public review):

#### Summary:

The authors integrate multiple large databases to test whether body sizes were positively associated with which species tolerate urban areas. In general, many plant families showed a positive association between body size and urban tolerance, whereas a smaller, though still non-trivial, percentage of animal families showed the same pattern. Notably, the authors are careful in the interpretation of their findings and provide helpful context for the ways that this analysis can be generative in shaping new hypotheses and theory around how urbanization influences biodiversity at large. They are careful to discuss how body size is an important trait, but the absence of a relationship between body size and urban tolerance in many families suggests a variety of other traits undergird urban success.

#### Strengths:

The authors aggregated a large dataset, but they also applied robust filters to ensure they had an adequate and representative number of detections for a given species, family, geography, etc. The authors also applied their analysis at multiple taxonomic scales (family and order), which allowed for a better interpretation of the patterns in the data and at what taxonomic scale body size might be important.

#### Weaknesses:

My main concern is that it is not fully clear how the measure of body size might influence the result. The authors were unable to obtain consistent measures of body size (mean, median, maximum, or sex variation). This, of course, could be very consequential as means and medians can differ quite a bit, and they certainly will differ substantially from a maximum. And of course, sex differences can be marked in multiple directions or absent altogether. The authors do note that they selected the measure that was most common in a family, but it was not clear whether species in that family that did not have that measure were removed or not. This could potentially shape the variability in the dataset and obscure true patterns. This may require additional clarity from the authors and is also a real constraint in compiling large data from disparate sources.

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### Reviewer #2 (Public review):

I have completed a thorough review of this paper, which seeks to use the large datasets of species occurrences available through GBIF to estimate variation in how large numbers of plant and animal species are associated with urbanization throughout the world, describing what they call the "species urbanness distribution" or SUD. They explore how these SUDs differ between regions and different taxonomic levels. They then calculate a measure of urban tolerance and seek to explore whether organism size predicts variation in tolerance among species and across regions.

The study is impressive in many respects. Over the course of several papers, Callaghan and coauthors have been leaders in using "big [biodiversity] data" to create metrics of how species' occurrence data are associated with urban environments, and in describing variation in urban tolerance among taxa and regions. This work has been creative, novel, and it has pushed the boundaries of understanding how urbanization affects a wide diversity of taxa. The current paper takes this to a new level by performing analyses on over 94000

observations from >30,000 species of plants and animals, across more than 370 plant and animal taxonomic families. All of these analyses were focused on answering two main questions:

- (1) What is the shape of species' urban tolerance distributions within regional communities?
- (2) Does body size consistently correlate with species' urban tolerance across taxonomic groups and biogeographic contexts?

Overall, I think the questions are interesting and important, the size and scope of the data and analyses are impressive, and this paper has a potentially large contribution to make in pushing forward urban macroecology specifically and urban ecology and evolution more generally.

Despite my enthusiasm for this paper and its potential impact, there are aspects that could be improved, and I believe the paper requires major revision.

Some of these revisions ideally involve being clearer about the methodology or arguments being made. In other cases, I think their metrics of urban tolerance are flawed and need to be rethought and recalculated, and some of the conclusions are inaccurate. I hope the authors will address these comments carefully and thoroughly. I recognize that there is no obligation for authors to make revisions. However, revising the paper along the lines of the comments made below would increase the impact of the paper and its clarity to a broad readership.

Major Comments:

#### (1) Subrealms

Where does the concept of "subrealms" come from? No citation is given, and it could be said that this sounds like an idea straight out of Middle Earth. How do subrealms relate to known bioclimatic designations like Koppen Climate classifications, which would arguably be more appropriate? Or are subrealms more socio-ecologically oriented? From what I can tell, each subrealm lumps together climatically diverse areas. It might be better and more tractable to break things in terms of continents, as the rationale for subrealms is unclear, and it makes the analyses and results more confusing. The authors rationalized the use of subrealms to account for potential intraspecific differences in species' response to urbanization, but that is never a core part of the questions or interpretation in the paper, and averaging across subrealms also accounts for intraspecific variation. Another issue with using the subrealm approach is that the authors only included a species if it had 100 observations in a given subrealm, leading to a focus on only the most common species, which may be biased in their SUD distribution. How many more species would be included if they did their analysis at the continental or global scale, and would this change the shape of SUDs?

#### (2) Methods - urban score

The authors describe their "urban score" as being calculated as "the mean of the distribution of VIIRS values as a relative species specific measure of a response to urban land cover."

I don't understand how this is a "relative species-specific measure". What is it relative to? Figures S4 and S5 show the mean distribution of VIIRS for various taxa, and this mean looks to be an absolute measure. Mean VIIRS for a given species would be fine and appropriate as an "urban score", but the authors then state in the next sentence: "this urban score represents the relative ranking of that species to other species in response to urban land cover".

That doesn't follow from the description of how this is calculated. Something is missing here. Please clarify and add an explicit equation for how the urban score is calculated because the text is unclear and confusing.

### (3) Methods - urban tolerance

How the authors are defining and calculating tolerance is unclear, confusing, and flawed in my opinion.

Tolerance is a common concept in ecology, evolution, and physiology, typically defined as the ability for an organism to maintain some measure of performance (e.g., fitness, growth, physiological homeostasis) in the presence versus absence of some stressor. As one example, in the herbivory literature, tolerance is often measured as the absolute or relative difference in fitness of plants that are damaged versus undamaged (e.g., <https://academic.oup.com/evolut/article/62/9/2429/6853425?login=true>).

On line 309, after describing the calculation of urban scores across subrealms, they write: "Therefore, a species could be represented across multiple subrealms with differing measures of urban tolerance (Fig. S4). Importantly, this continuous metric of urban tolerance is a relative measure of a species' preference, or affinity, to urban areas: it should be interpreted only within each subrealm".

This is problematic on several fronts. First, the authors never define what they mean by the term "tolerance". Second, they refer to urban tolerance throughout the paper, but don't describe the calculation until lines 315-319, where they write (text in [ ] is from the reviewer):

"Within each subrealm, we further accounted for the potential of different levels of urbanization by scaling each species' urban score by subtracting the mean VIIRS of all observations in the subrealm (this value is hereafter referred to as urban tolerance). This 'urban tolerance' (Fig. S5) value can be negative - when species under-occupy urban areas [relative to the average across all species] suggesting they actively avoid them-or positive-when species over-occupy urban areas [relative to the average across all species] suggesting they prefer them (i.e., ranging from urban avoiders to urban exploiters, respectively).

They are taking a relativized urban score and then subtracting the mean VIIRS of all observations across species in a subrealm. How exactly one interprets the magnitude isn't clear and they admit this metric is "not interpretative across subrealms".

This is not a true measure of tolerance, at least not in the conventional sense of how tolerance is typically defined. The problem is that a species distribution isn't being compared to some metric of urbanness, but instead it is relative to other species' urban scores, where species may, on average, be highly urban or highly nonurban in their distribution, and this may vary from subrealm to subrealm. A measure of urban tolerance should be independent of how other species are responding, and should be interpretable across subrealms, continents, and the globe.

I propose the authors use one of two metrics of urban tolerance:

(i) Absolute Urban Tolerance = Mean VIIRS of species<sub>i</sub> - Mean VIIRS of city centers

Here, the mean VIIRS of city centers could be taken from the center of multiple cities throughout a subrealm, across a continent, or across the world. Here, the units are in the original VIIRS units where 0 would correspond to species being centered on the most extreme urban habitats, and the most extreme negative values would correspond to species that occupy the most non-urban habitats (i.e., no artificial light at night). In essence, this measure of tolerance would quantify how far a species' distribution is shifted relative to the most highly urbanized habitat available.

(ii) % Urban Tolerance = (Mean VIIRS of species<sub>i</sub> - Mean VIIRS of city centers)/Mean VIIRS of city centers \* 100%

This metric provides a % change in species mean VIIRS distribution relative to the most urban habitats. This value could theoretically be negative or positive, but will typically be negative, with -100% being completely non-urban, and 0% being completely urban tolerant.

Both of these metrics can be compared across the world, as it would provide either absolute (equation 1) or relative (equation 2) metrics of urban tolerance that are comparable and easily interpretable in any region.

In summary, the definition of tolerance should be clear, the metric should be a true measure of tolerance that is comparable across regions, and an equation should be given.

(4) Figure 1: The figure does not stand alone. For example, what is the hypothesis for thermophily or the temperature-size rule? The authors should expand the legend slightly to make the hypotheses being illustrated clearer.

(5) SUDs: I don't agree with the conclusion given on line 83 ("pattern was consistent across subrealms and several taxonomic levels") or in the legend of Figure 2 ("there were consistent patterns for kingdoms, classes, and orders, as shown by generally similar density histograms shapes for each of these").

The shapes of the curves are quite different, especially for the two Kingdoms and the different classes. I agree they are relatively consistent for the different taxonomic Orders of insects.

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### Reviewer #3 (Public review):

Summary:

This paper reports on an association between body size and the occurrence of species in cities, which is quantified using an 'urban score' that can be visualized as a 'Species Urbanness Distribution' for particular taxa. The authors use species records from the Global Biodiversity Information Facility (GBIF) and link the occurrence data to nighttime lighting quantified using satellite data (Visible Infrared Imaging Radiometer Suite-VIIRS). They link the urban score to body size data to find 'heterogeneous relationship between body size and urban tolerance across the tree'. The results are then discussed with reference to potential mechanisms that could possibly produce the observed effects (cf. Figure 1).

Strengths:

The novelty of this study lies in the huge number of species analyzed and the comparison of results among animal taxa, rather than in a thorough analysis of what traits allow species to persist under urban conditions. Such analyses have been done using a much more thorough approach that employs presence-absence data as well as a suite of traits by other studies, for example, in (Hahs et al. 2023, Neate-Clegg et al. 2023). The dataset that the authors produced would also be very valuable if these raw data were published, both the cleaned species records as well as the body sizes.

The paper could strongly add to our understanding of what species occur in cities when the open questions are addressed.

Weaknesses:

I value the approach of the authors, but I think the paper needs to be revised.

In my view, the authors could more carefully validate their approach. Currently, any weakness or biases in the approach are quickly explained away rather than carefully explored. This concerns particularly the use of presence-only data, but also the calculation of the urban score.

The vast majority of data in GBIF is presence-only data. This produces a strong bias in the analysis presented in the paper. For some taxa, it is likely that occurrences within the city are overrepresented, and for other taxa, the opposite is true (cf. Sweet et al. 2022). I think the authors should try to address this problem.

The authors should compare their results to studies focusing on particular taxa where extensive trait-based analyses have already been performed, i.e., plants and birds. In fact, I strongly suggest that the authors should compare their results to previous studies on the relationship between traits, including body size and occurrences along a gradient of urbanisation, to draw conclusions about the validity of the approach used in the current study, which has a number of weaknesses.

They should be more careful in coming up with post-hoc explanations of why the pattern found in this study makes sense or suggests a particular mechanism. This reviewer considers that there is no way in which the current study can disentangle the different possible mechanisms without further analyses and data, so I would suggest pointing out carefully how the mechanisms could be studied

More details should be given about the methodology. The readers should be able to understand the methods without having to read a number of other papers.

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